Joint Aeromagnetic Compensation and Magnetic-Anomaly Navigation with a Fixed-Lag Factor Graph

Jayden Dongwoo Lee
Korea Advanced Institute of Science and Technology, Daejeon, Korea

Bohyun Chang
Korea Advanced Institute of Science and Technology, Daejeon, Korea

Hyochoong Bang, Member, IEEE
Korea Advanced Institute of Science and Technology, Daejeon, Korea

***Abstract*— Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft's own field must be compensated online when no calibration flight is available, the cold-start regime. A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We instead cast navigation and compensation as one maximum-a-posteriori problem on a factor graph, carrying the TL coefficients as time-varying graph variables so the compensation adapts along the flight with no neural network. That single problem is solved across a spectrum of lags under one cost function: a full-line batch that sets the offline accuracy bound, a fixed-lag sliding window, and a per-epoch incremental smoother that reaches the real-time limit at $7.8\,\mathrm{ms}$ per state. The lag is the primary design variable, trading the future data that corrects a committed state against latency. A linear condition states when position can be separated from compensation and predicts a cold-start transient that the experiments decompose into a warm-prior part and a map-observability part. Across cold-start lines from two flights and two maps, the smoother is the most accurate of four estimators on every full-length case (8 of 8): it stays bounded where an online-TL EKF diverges and a marginalized particle filter is erratic, and, carrying no learned component of its own, it beats a tuned EKF+TL+NN baseline. At the shortest lag the compensation prior decides the cold start. One scalar standard deviation shared by a regressor whose columns span four orders of magnitude leaves the first linearizations stiff in the compensation relative to position, so roughly $2000\,\mathrm{nT}$ of uncompensated cabin field is charged to position through the map gradient and the estimate leaves the map. No lag, robust kernel, map resolution, relinearization schedule, measurement rate, process-noise floor, or single choice of measurement-noise scale repairs it; raising that one standard deviation by two orders of magnitude removes it on every affected line. A Monte-Carlo simulation confirms the reported covariance is consistent, and an independent GTSAM reimplementation sharing no code reproduces the bounded result to within a metre on both magnetometers once the compensation cold start is handled.**

***Index Terms*— Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.**

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J. D. Lee, B. Chang, and H. Bang are with the Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, Daejeon 34141, Republic of Korea (e-mail: cin6474@kaist.ac.kr; hcbang@kaist.ac.kr).

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Nomenclature

$\mathbf{x}_t$	INS error state at epoch t (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$.
$\boldsymbol{\beta}_t$	Online Tolles–Lawson compensation coefficients.
$\boldsymbol{\chi}_t$	Joint state $(\mathbf{x}_t, \boldsymbol{\beta}_t)$.
$\boldsymbol{\Phi}_t, \mathbf{Q}_t$	Pinson state-transition matrix and process covariance.
z_t, η_t	Scalar magnetometer measurement and its noise, variance R.
$h(\cdot)$	Anomaly-map interpolation (with IGRF core) as a function of position.
$\mathbf{g}_t$	Map gradient $\partial h/\partial \mathbf{p}$ at epoch t.
$\mathbf{A}_t$	Tolles–Lawson basis row from the three-axis fluxgate.
c_t	Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$.
L_w, L_o	Sliding-window length and overlap (look-ahead).
ρ, δ	Robust kernel and its Huber threshold.
$\mathbf{G}, \boldsymbol{\Psi}$	Stacked map-gradient and TL-regressor matrices over a window.
$\mathcal{N}(\cdot,\cdot)$	Gaussian density.
$\lVert\cdot\rVert_{\mathbf{M}}$	Mahalanobis norm, $\lVert\mathbf{v}\rVert_{\mathbf{M}}^2 = \mathbf{v}^\top\mathbf{M}\mathbf{v}$.

Boldface lowercase denotes vectors and boldface uppercase matrices.

I. Introduction

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, and observable with a lightweight scalar magnetometer [3], [4]. It is also difficult to spoof from a stand-off distance. Denial of a satellite signal exploits the far-field radio link, whose power falls off only as the inverse square of range, so a modest transmitter jams or spoofs a receiver from far away; a magnetic anomaly, by contrast, is a near-field source whose dipole field decays as the inverse cube of distance, so counterfeiting a credible anomaly at a moving aircraft would demand an impractically large or close field source [5]. The same map-matching idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [6], [7], [8]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft's own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The idea has matured from feasibility study to flight demonstration: recursive Bayesian map matching, from the extended Kalman filter to grid and particle estimators, was characterized on flight data [3], [4] and flown on an F-16 [9], and recent multi-hour fixed-wing trials over thousands of kilometres report metre-to-decametre accuracy with quantum magnetometers [5]. As the estimators have matured, attention has moved to the inputs they rely on, and a consensus is forming that the pacing requirement for operational deployment is a sufficiently accurate, uncertainty-aware anomaly map [10]. Two obstacles, in fact, separate these demonstrations from routine use, and they are distinct: the reference map on the ground and the aircraft's own field in the air. The map is a survey-and-data problem, pursued elsewhere [10]; this paper addresses the second.

The platform's own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position. This aircraft-field problem is intrinsic to MagNav and independent of any particular airframe; it is most acute for magnetometers mounted inside the cabin, close to avionics and current-carrying structure, rather than on a wingtip stinger boom that holds the sensor away from the platform field.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [11], [12], [13], a regression of the interference onto sixteen attitude-dependent terms (permanent, induced, and eddy-current) formed from a three-axis fluxgate magnetometer. It faces two difficulties that are easy to conflate but are in fact orthogonal. The first is temporal: the coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, yet many operational scenarios admit none, and the coefficients drift with temperature and configuration. In this cold-start regime they must be estimated online and jointly with navigation, from the same measurements that fix position, a coupling that makes both estimation and its analysis substantially harder than the calibrated case. The second is representational: however well its coefficients are fit, the linear TL basis cannot span interference from platform currents and other non-rigid, non-attitude sources, which contribute a large fraction of the field on cabin-mounted sensors [14], [15]. One difficulty is about identifying a known model class in time; the other is about that class being too small.

A public benchmark has made both concrete: the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset [14], with flight-grade INS, wingtip-stinger and cabin-mounted magnetometers, and high-resolution anomaly maps. Most work on it attacks the two difficulties offline: the TL coefficients by a batch calibration flight, and the representational gap by learned models fit offline on logged data, neural-network heading-error and denoising compensators [16], [15], physics-informed liquid-time-constant networks [17], and model-stability studies of the regression in gradiometry [18]. Offline learning, however, presupposes exactly what the operational scenario lacks: a corpus of logged flights from the same aircraft, a reference-grade compensated sensor flying alongside to supply training targets, the computation to train on that corpus, and a willingness to retrain whenever the platform configuration or its magnetic state drifts. A compensator fit this way is specific to one airframe in one equipment state, and its accuracy in flight is bounded by how well the training corpus transfers to the day's conditions. We therefore treat the offline family as context rather than competition: every estimator compared in this paper starts, as ours does, from no prior flight data. The exception is the online NN-in-EKF line of work originated by Gnadt [15], which carries the TL coefficients and the weights of a residual network as filter states learned in flight, and which has recently been extended with an error-state formulation and a natural-gradient interpretation [19]; this lineage holds, to our knowledge, the strongest reported cold-start results on the benchmark. This paper takes on the temporal difficulty fully: online, joint estimation of a time-varying compensation. The representational gap it does not model term-by-term: instead of a learned remainder, the estimator absorbs what the TL basis misses structurally, through the time-varying coefficients, a colored disturbance state, and a robust kernel, and Section V verifies against the recorded platform currents how much of the remainder this accounts for.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [20] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [6] and nonlinear Kalman filtering for terrain-aided navigation [7] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [21], [22], [23], [8]. For MagNav specifically, Canciani and Raquet combined a

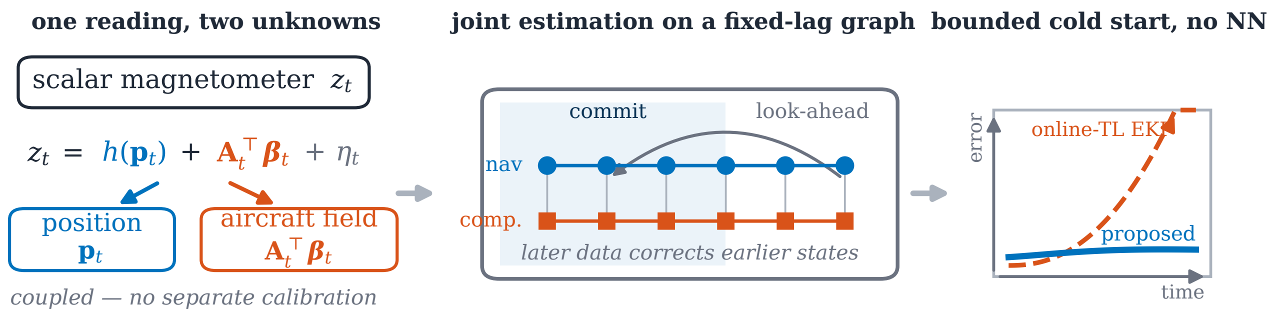

Fig. 1. Concept of the study: (a) one scalar reading couples aircraft position and aircraft-field compensation; (b) a fixed-lag factor graph estimates both jointly, letting later measurements correct earlier states; (c) bounded cold-start navigation, with no neural network.

calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [9]. Beyond the EKF, unscented filtering [24], invariant formulations that respect the state manifold [25], and mixture/particle representations [26] improve consistency under nonlinearity. However, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and, unable to revise a committed state, the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [11], [12], [13]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [16], model-stability and robustness of the regression have been studied in gradiometry [18], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [15]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses, introduced with Gnadt's online models [15] and recently extended in [19]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and holds, to our knowledge, the strongest reported cold-start results on that benchmark. Nevertheless, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and, being embedded in an EKF, inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [27] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [28], [29], the same structure underlying bundle adjustment and graph SLAM [30], [31], [32]. Incremental and square-root smoothers (square-root SAM [33], iSAM [34], iSAM2 [35], and exactly-sparse delayed-state filters [36]) make the solution efficient enough for online use, and fixed-lag sliding-window optimization [37], [38] bounds its cost. On-manifold inertial preintegration [39], factor-graph inertial fusion [40], and modern visual–inertial estimators [41] have brought the approach firmly into navigation. Despite these advances, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks. The factor graph has very recently reached aeromagnetic compensation itself: Lathrop et al. [42] calibrate a magnetometer by fusing vector and scalar magnetometers with inertial measurements in a factor graph, absorbing large permanent moments and a time-varying external field. That work compensates the sensor but does not close the loop to navigation. The joint problem, estimating a time-varying compensation together with aircraft position from map matching, has not, to our knowledge, been posed in a published factor-graph formulation, and the observability of doing so has not been characterized.

Taken together, the three lines leave a specific gap. The cold-start MagNav problem is fundamentally a joint estimation of position and a time-varying compensation model from a single scalar stream, in which the two unknowns are confounded through the very measurement that must resolve them. Causal filters (cluster A) must commit each state before the maneuvers that would disambiguate the compensation have occurred, and so diverge when the initial residual is large; learned online compensation (cluster B) repairs that residual but keeps the causal, commit-online architecture and adds parameters that are difficult to certify; and factor-graph smoothing (cluster C), which can defer commitment and estimate structured nuisance variables jointly, has not been applied to this problem, nor has anyone characterized when the joint problem is even observable. What is missing is an estimator that (i) defers commitment so that late-arriving calibration information corrects early navigation states, (ii) represents the compensation as first-class, time-varying variables rather than a black box, and (iii) comes with an explicit condition for when position and compensation can be separated at all. This paper supplies all three.

In this work we cast magnetic-anomaly navigation as a factor graph and, within it, carry the online Tolles–Lawson coefficients as time-varying variables, so that navigation and compensation are solved jointly as one maximum-a-posteriori (MAP) problem (Fig. 1). Rather than fix a single filter or smoother, we solve that one problem across a spectrum of lags: a full-line batch that uses every measurement and sets the offline accuracy bound, a fixed-lag sliding window that commits a state once it has seen a bounded amount of future data, and a per-epoch incremental smoother, built on iSAM2 [35], that takes the lag to its real-time limit. The lag is the primary design variable along this spectrum, trading observability, the amount of future data that corrects a committed state, against latency and computation. What unites every point on it, and separates all of them from the causal filters of clusters A and B, is that each committed state can be relinearized and revised as later measurements arrive, so late calibration corrects early navigation; a causal filter commits each state once and cannot revise it. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [43], so smoothing per se is not the contribution; the joint estimation of a time-varying compensation, realized across the lag spectrum, is. The paper makes the following contributions:

1) **Factor-graph joint compensation.** We carry the online Tolles–Lawson coefficients as time-varying variables of a factor graph so that navigation and compensation are estimated jointly from the same scalar measurements. To the authors' knowledge this is the first factor-graph formulation that jointly estimates time-varying Tolles–Lawson compensation coefficients and navigation states in the calibration-cold-start regime. Relinearizing the coefficients as the graph is re-solved makes the compensation adapt along the flight, the role a learned online model plays in the competing filter, while the graph keeps every quantity an explicit, physically interpretable state.
2) **A MAP estimation spectrum.** We solve that one MAP problem across a spectrum of lags under a single cost function: an offline full-line batch, a fixed-lag sliding window, and a per-epoch incremental smoother built on iSAM2. The lag is the primary design variable trading observability against latency and computation, so the batch fixes the offline accuracy bound and the incremental smoother runs in real time ($6.0\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state at a $300\,\mathrm{s}$ lag and $1\,\mathrm{Hz}$ states, two orders of magnitude within the state cadence). This places recursive filtering and batch smoothing as two limits of one estimator rather than competing methods, and every point on the spectrum can re-smooth the past, the property the cold start needs.
3) **Position–compensation separability and the cold-start transient.** We give a linear-algebraic condition, evaluated over a window, for when a position error can be distinguished from a change in the compensation, separating three failure modes: an unexcited trajectory, a rank-deficient compensation basis, and the subtle one, a compensation subspace aligned with the anomaly gradient. The condition predicts the cold-start transient, which the experiments decompose: a warm compensation prior cuts it by $40\,\%$, and the remaining transient tracks map-gradient observability rather than compensation error. Separately, and ahead of any observability question, we identify the setting that decides whether a short-lag smoother survives a cold start at all: the scale of the compensation prior relative to the regressor. We measure the failure, rule out the lag, the robust kernel, the map resolution, the relinearization schedule, the measurement rate, the measurement-noise scale and the process-noise floor as causes, and show that raising the compensation prior by two orders of magnitude, or setting it per regressor column, removes it on every affected line.
4) **Flight-data validation.** On uncompensated cabin magnetometers across eight cold-start cases spanning four lines, two flights, and two maps, the smoother is the most accurate of four cold-start estimators. Run rather than merely cited, the online EKF+TL+NN lineage of [15], [19] is kept bounded by its network where a weak online-TL EKF diverges and a marginalized particle filter is erratic, yet the smoother beats it with no learned component. A Monte-Carlo simulation confirms the reported covariance is consistent, and an independent GTSAM reimplementation [29] that shares no code reproduces the bounded cold-start result, separating the estimator from its implementation.

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the factor graph and solves it across the lag spectrum, from the batch bound to the per-epoch incremental smoother. Section IV establishes the separability condition. Section V describes the baselines and reports and interprets the experiments. Section VI concludes and states the limitations.

II. Problem Formulation

The situation this paper addresses is the cold start: an aircraft departs with an uncalibrated magnetometer installation, its INS error grows, and the only absolute position information is a scalar magnetic reading in which the map signal and the aircraft's own field are mixed. The estimation task is therefore joint from the outset: there is no separate calibration step whose output a navigator could consume:

Problem 1 (Joint compensation and navigation): Given the scalar measurements $z_{1:N}$, the fluxgate regressor rows $\mathbf{A}_{1:N}$, the anomaly map h, the INS mechanization, and the prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$, estimate the joint trajectory $\boldsymbol{\chi}_{1:N}$ of navigation errors and compensation coefficients, $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$.

The rest of the section supplies the three models that make Problem 1 concrete: how the navigation error evolves, what the scalar magnetometer measures, and how the aircraft field enters it. Section III then casts the joint estimation as maximum-a-posteriori inference on a factor graph and solves it across a spectrum of lags.

A. INS error state and dynamics

We adopt the standard Pinson error model of a strapdown INS [1], [2], [44]. The error state

$$\mathbf{x} = \left[\, \delta\mathbf{p}^\top \; \delta\mathbf{v}^\top \; \boldsymbol{\psi}^\top \; h_a \; \hat{a} \; \mathbf{b}_a^\top \; \mathbf{b}_g^\top \; S \,\right]^\top \in \mathbb{R}^{18} \tag{1}$$

stacks position error $\delta\mathbf{p}$, velocity error $\delta\mathbf{v}$, attitude error $\boldsymbol{\psi}$ (three components each), the baro-inertial altitude-aiding pair $(h_a, \hat{a})$ that damps the vertically unstable channel, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. The discretized, linearized dynamics are

$$\mathbf{x}_{t+1} = \boldsymbol{\Phi}_t \mathbf{x}_t + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \tag{2}$$

where $\boldsymbol{\Phi}_t$ is the Pinson state-transition matrix (Appendix A) and $\mathbf{Q}_t$ the process covariance; the FOGM block of $\boldsymbol{\Phi}_t$ is $e^{-\Delta t/\tau}$ with correlation time τ.

B. Scalar magnetometer measurement

The scalar magnetometer reads

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R), \tag{3}$$

where $h(\cdot)$ interpolates the crustal anomaly map (with the IGRF core field restored) at the aircraft position $\mathbf{p}_t = \bar{\mathbf{p}}_t + \delta\mathbf{p}_t$ and c_t is the Tolles–Lawson part of the aircraft interference; the non-Tolles–Lawson remainder is carried by the first-order Gauss–Markov state S_t introduced below. Linearizing about the nominal position,

$$z_t \approx h(\bar{\mathbf{p}}_t) + \mathbf{g}_t^\top \delta\mathbf{p}_t + c_t + \eta_t, \qquad \mathbf{g}_t = \left.\frac{\partial h}{\partial \mathbf{p}}\right|_{\bar{\mathbf{p}}_t}, \tag{4}$$

so the map gradient $\mathbf{g}_t$ is the sole source of horizontal position information: where $\mathbf{g}_t \approx \mathbf{0}$ (a locally flat map) position is momentarily unobservable, and everywhere the interference c_t competes with the $\mathbf{g}_t^\top \delta\mathbf{p}_t$ term for the same measurement. The map h is stored as a gridded anomaly product upward-continued to the survey altitude; we evaluate it and its gradient by bicubic interpolation, which yields the continuous $\mathbf{g}_t$ required to linearize (4), and we restore the IGRF core field so that z_t and h are on a common absolute scale. The gradient magnitude $\|\mathbf{g}_t\|$ varies by more than an order of magnitude along a line, which is why position accuracy is not uniform in time even for a perfect compensator, a point the error traces of Section V make visible.

C. Tolles–Lawson interference model

Classical aeromagnetic compensation writes the interference as a linear combination of attitude-dependent basis terms [11], [12], [13],

$$c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t, \tag{5}$$

where $\mathbf{A}_t \in \mathbb{R}^m$ is the TL basis row of (5) built from the three-axis fluxgate; its classical rank deficiency is a gauge freedom (Remark 1). With direction cosines $\hat{\mathbf{b}}_t = [\, u \; v \; w \,]^\top$ of the Earth field in body axes and total field magnitude B_t, the permanent (3), induced (6), and eddy-current (9) groups are

$$\mathbf{A}_t = \big[\, \underbrace{u, v, w}_{\text{perm.}}, \; \underbrace{B_t u^2, \ldots, B_t w^2}_{\text{induced}}, \; \underbrace{B_t u\dot{u}, \ldots, B_t w\dot{w}}_{\text{eddy}} \,\big]^\top, \tag{6}$$

optionally augmented by a bias term, giving $m \le 19$ coefficients $\boldsymbol{\beta}_t$. In the calibrated setting $\boldsymbol{\beta}$ is fit once and frozen; in the cold-start setting it is unknown at takeoff and drifts slowly, so we treat it as a time-varying state with a random-walk prior $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{w}_t^\beta$, $\mathbf{w}_t^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta)$.

III. MAP Estimation Spectrum

Problem 1 has a single maximum-a-posteriori solution; what a designer chooses is how much of the flight to weigh before committing an estimate. Solving it over a lag of L epochs traces a spectrum under one cost function: the full-line batch ($L = N$) uses every measurement and is the offline accuracy bound; a per-epoch recursion (small L) commits each state as soon as it is measured, for real-time use; and the fixed-lag sliding window sits in between, committing a state once it has seen a bounded amount of future data. The lag is thus the primary design variable that trades observability, through the future data each committed estimate sees, against latency and computation, and every point on the spectrum solves the same factor graph. This section develops that graph and its batch solution, then the fixed-lag window, then the per-epoch incremental smoother that is the window's real-time limit, and finally the robust kernel and sparse solver common to all three; the estimator is drawn in Fig. 2, and the three realizations of the spectrum are sketched in Fig. 3.

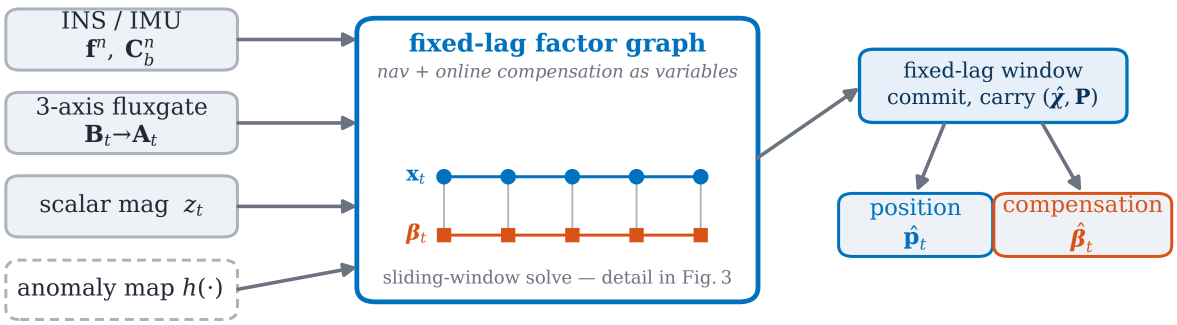

Fig. 2. Proposed estimator: the sensors feed a factor graph of navigation, online TL compensation, solved over a fixed-lag window.

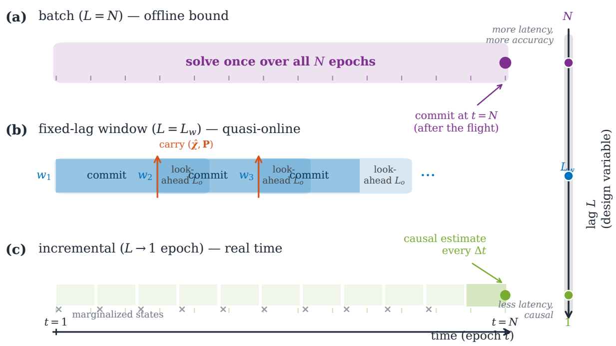

Fig. 3. The MAP estimation spectrum: the same factor graph solved at three lags, from the full-line batch bound through the fixed-lag window to the per-epoch incremental smoother, trading the future data each committed estimate sees against latency.

A. Batch MAP estimation (offline bound)

The three models of Section II couple through a single scalar stream: the same measurement z_t must simultaneously fix position through the map term $h(\mathbf{p}_t)$ and identify the compensation $\mathbf{A}_t^\top \boldsymbol{\beta}_t$. We take the maximum-a-posteriori (MAP) estimate of Problem 1. Because the dynamics are Markov and each measurement depends only on the current state, the posterior splits into one factor per piece of information: a prior, one dynamics factor per step for each of the two chains, and one measurement factor per epoch,

$$p(\boldsymbol{\chi}_{1:N} \mid z_{1:N}) \;\propto\; p(\boldsymbol{\chi}_1) \prod_{t=1}^{N-1} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)\, p(\boldsymbol{\beta}_{t+1} \mid \boldsymbol{\beta}_t) \times \prod_{t=1}^{N} p(z_t \mid \boldsymbol{\chi}_t). \tag{7}$$

This factorization is a factor graph [27], [28]: a bipartite graph whose variable nodes carry the unknowns and whose factor nodes carry the individual densities in (7), each edge joining a factor to the variables it constrains (Fig. 4). Concretely, there is one variable node per epoch holding the joint state $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$, the 18 Pinson error states of (1) and the $m \le 19$ TL coefficients of (6), so the graph has N variable nodes and $N\,(18{+}m)$ scalar unknowns. Four factor types populate it: a single unary prior factor on $\boldsymbol{\chi}_1$; $N{-}1$ binary INS dynamics factors, each linking $\mathbf{x}_t$ to $\mathbf{x}_{t+1}$ through (2); $N{-}1$ binary compensation random-walk factors, each linking $\boldsymbol{\beta}_t$ to $\boldsymbol{\beta}_{t+1}$; and N unary measurement factors, each attached to one node $\boldsymbol{\chi}_t$ through (3). The two dynamics families run as parallel chains that never touch directly; the measurement factor is the only place position and compensation meet, which is exactly why the two are estimated jointly and why their separability needs a condition. MAP estimation on such a graph is sparse nonlinear least squares: the negative logarithm of (7) turns each factor into a squared or robust residual, and the Gauss–Newton normal equations inherit the graph's chain sparsity, which is what makes batch solution tractable at flight scale. Concretely, the MAP estimate minimizes

$$J(\boldsymbol{\chi}) = \|\boldsymbol{\chi}_1 - \boldsymbol{\chi}_0\|^2_{\mathbf{P}_0^{-1}} + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \boldsymbol{\Phi}_t \mathbf{x}_t\|^2_{\mathbf{Q}_t^{-1}} + \sum_{t=1}^{N-1} \|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}} + \sum_{t=1}^{N} \rho\!\left(\frac{r_t(\boldsymbol{\chi}_t)}{\sqrt{R}}\right), \tag{8}$$

Its four terms are, in order, the prior, the INS dynamics, the compensation random walk, and the measurement misfit under a possibly robust kernel ρ; the measurement residual $r_t(\boldsymbol{\chi}_t)$ of (9) is what the model cannot yet explain. The prior anchors both blocks of $\boldsymbol{\chi}_1 = (\mathbf{x}_1, \boldsymbol{\beta}_1)$. Its $\boldsymbol{\beta}$-block $\mathbf{P}_0^\beta$ is the one setting a cold start is sensitive to, because it decides how far the compensation may travel from zero before the navigation states start absorbing the platform field; we return to it in Section H and report its value in Table I. We make the standard modeling assumptions of aided inertial navigation [1], [26].

Assumption 1:
The process and measurement noises $\mathbf{w}_t, \mathbf{w}_t^\beta, \eta_t$ are zero-mean, white, mutually independent, and Gaussian; the linearizations (2) and (4) are valid over a window; and the prior $(\hat{\chi}_0, \mathbf{P}_0)$ is Gaussian.

Assumption 1 is the same one under which the EKF and RTS smoother are derived, so any performance difference between the estimators reflects their structure, not their noise models. What Problem 1 adds over classical map-matching is exactly the second chain: navigation and compensation are estimated from the same evidence, and Section IV characterizes when that joint problem is well-posed.

With the compensation coefficients $\boldsymbol{\beta}$ carried as graph variables throughout $\boldsymbol{\chi}_t$, the measurement factor of (3) is, in residual form,

$$r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t, \tag{9}$$

with Gaussian information R^{-1}, and its Jacobian row is

$$\mathbf{H}_t = \Big[\, \underbrace{\mathbf{g}_t^\top}_{\delta\mathbf{p}} \quad \underbrace{\mathbf{0}}_{\delta\mathbf{v},\boldsymbol{\psi},h_a,\hat{a},\mathbf{b}} \quad \underbrace{\mathbf{A}_t^\top}_{\boldsymbol{\beta}} \quad \underbrace{1}_{S} \,\Big]. \tag{10}$$

These robust scalar-magnetometer factors, together with the Pinson–FOGM process factors, the compensation random-walk factors, and the joint prior on $\boldsymbol{\chi}_1$, form the four factor families of the graph in Fig. 4. The random-walk factor sets how quickly the coefficients may adapt, with $\mathbf{Q}^\beta \to \mathbf{0}$ recovering a single static $\boldsymbol{\beta}$ over the line. The INS mechanization, the fluxgate (through $\mathbf{A}_t$), and the anomaly map (through h and $\mathbf{g}_t$) enter as known factor parameters, not as estimated variables. Two implementation details: in the online estimator only the horizontal components of $\mathbf{g}_t$ are used (the vertical channel is baro-damped and the map is evaluated at survey altitude), whereas the batch validation of Section V-A uses the full three-component gradient; and the implementation orders the state vector as [Pinson-17; $\boldsymbol{\beta}$; S], a permutation of (1) that does not affect the estimate. Because consecutive coefficient blocks are chained rather than instantaneous, the compensation is identified from the whole trajectory, and because the batch solution uses future as well as past measurements, early navigation states benefit from calibration information that only becomes observable after later maneuvers, the property a causal filter lacks.

Absorbing the non-Tolles–Lawson remainder. On this dataset the cabin interference is dominated by platform currents and other sources the attitude-only TL basis $\mathbf{A}_t$ cannot span, and the online EKF+TL+NN lineage [15], [19] devotes a neural network precisely to this remainder. The proposed graph carries no network; instead three of the factors above combine to absorb the remainder as a structured, robust regression problem. The random-walk factor makes $\boldsymbol{\beta}_t$ time-varying: writing the true interference as $\mathbf{A}_t^\top \boldsymbol{\beta}^\star + e_t$, a static fit ($\mathbf{Q}^\beta \to \mathbf{0}$) forces the entire current signature into the residual e_t, whereas a nonzero $\mathbf{Q}^\beta$ lets the estimated coefficients track the slowly drifting, maneuver-correlated projection of the currents onto $\mathbf{A}_t$, the low-frequency part a single calibration fit cannot recover. The disturbance state S_t in the residual (9), a first-order Gauss–Markov process folded into the Pinson–FOGM process factor, then carries the temporally correlated component of e_t that is orthogonal to $\mathbf{A}_t$; it is the linear–Gaussian, single-scalar analogue of the learned residual, cheaper and far less expressive but requiring no training data. What neither term explains, the heavy-tailed switching transients of onboard electronics, is handled not by fitting but by the robust kernel of the measurement factor, which the reweighting of Section III-D down-weights so that such samples cannot pull the position estimate. The separability condition of Section IV speaks to this: where position and compensation are separable, S_t and $\boldsymbol{\beta}_t$ absorb only signal that is not collinear with the map gradient $\mathbf{g}_t$, so genuine map information is not silently explained away as interference. The trade it makes, a higher compensation residual than the network in exchange for navigation robustness, is quantified with the cold-start results in Section V-D, where the account is also checked against the recorded platform-current channels.

B. Fixed-lag sliding window

A single batch over a long line with a static $\boldsymbol{\beta}$ underfits the time-varying interference; carrying $\boldsymbol{\beta}$ as a time-varying chain over the full line removes that limitation but at a computation and latency that grow without bound, while a per-epoch filter, conversely, cannot use future data. We take the middle path and process the flight in overlapping windows of length L_w with overlap L_o, each a batch factor graph, in the manner of fixed-lag and incremental smoothing [35], [37], [38]. Window k spans epochs $[s_k, s_k + L_w)$; it is solved as a batch and commits only its leading stride of $L_w - L_o$ epochs. When the window slides, the states that leave it are marginalized out and their information is carried forward as a Gaussian prior on the retained states of window $k{+}1$ (Fig. 5), the standard fixed-lag marginalization of square-root SAM and sliding-window visual–inertial estimators [33], [35], [38], [41]. Because the graph is a chain, the marginal of the departing sub-graph reduces to the forward-filtered state at the commit boundary (a Schur complement computed for free by the forward Kalman pass), so each overlap measurement enters the estimate exactly once. The trailing overlap L_o thus acts as bounded smoother look-ahead, giving each committed estimate L_o epochs of future data, while the marginalized handoff keeps both the mean and the covariance consistent across window boundaries. Algorithm 1 summarizes the procedure; its per-window cost is constant in the line length, giving overall linear complexity.

It is worth being precise about the causal failure this repairs, because the same time-varying model (5) can be, and is, carried inside an EKF. At cold start the coefficient error makes the residual hundreds of nanotesla, and a one-pass filter must apportion that residual between position and $\boldsymbol{\beta}$ immediately, using only its current covariance, before the maneuvers that would separate the two have occurred. A wrong apportionment moves the position estimate, and the next update then evaluates the map gradient $\mathbf{g}_t$ at the wrong place, so even a correct residual is charged to the wrong direction: the linearization point

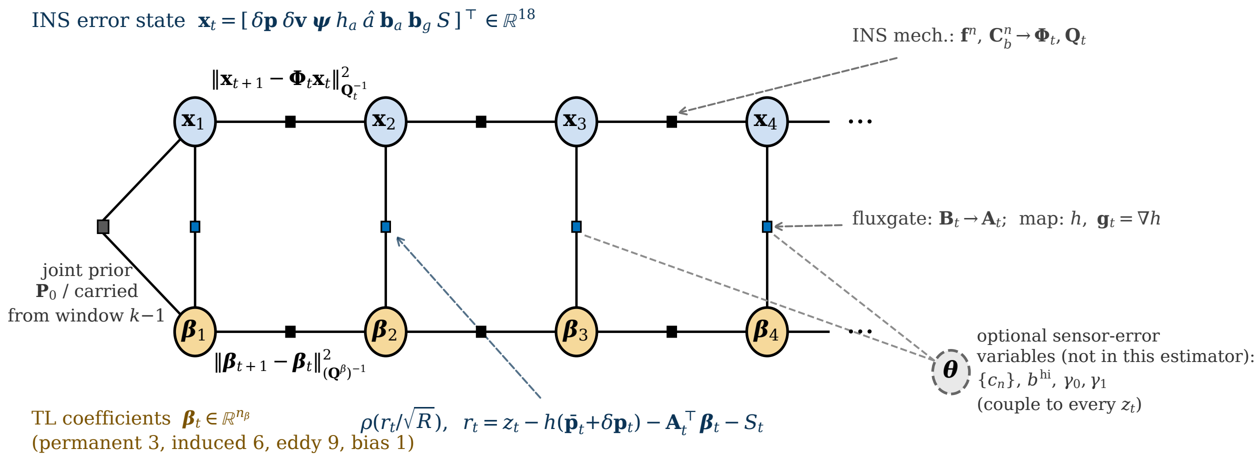

Fig. 4. Factor graph of one window: the 18-state INS error chain $\mathbf{x}_t$ and the time-varying TL chain $\boldsymbol{\beta}_t$, coupled by robust scalar-magnetometer factors. Dashed arrows are known factor parameters; the dashed node $\boldsymbol{\theta}$ holds optional extension variables.

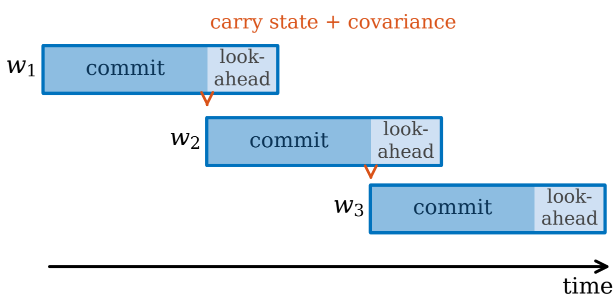

Fig. 5. Fixed-lag sliding window: each window commits its leading stride and carries the smoothed state and covariance to the next; the overlap is the look-ahead.

Algorithm 1 Fixed-lag sliding-window smoother

Input: measurements $\{z_t\}$, TL rows $\{\mathbf{A}_t\}$, map h, window L_w, overlap L_o, prior $(\hat{\chi}_0, \mathbf{P}_0)$
$s \leftarrow 1$
while $s \leq T$ **do**
  build factor graph on epochs $[s, \min(s{+}L_w, T))$ with prior $(\hat{\chi}_{s-1}, \mathbf{P}_{s-1})$
  repeat ▷ robust Gauss–Newton
    linearize residuals (9); form weights w_t from ρ (12)
    solve the linearized chain by an iterated RTS pass [43] (equivalently, sparse QR / square-root SAM [33]; Sec. III-D)
    update $\chi \leftarrow \chi + \Delta\chi$
  until converged
  commit epochs $[s, s{+}L_w{-}L_o)$; recover $(\hat{\chi}, \mathbf{P})$ at epoch $s{+}L_w{-}L_o{-}1$
  $s \leftarrow s + (L_w - L_o)$
end while
Output: committed trajectory $\{\hat{\mathbf{p}}_t, \hat{\boldsymbol{\beta}}_t\}$

and the estimate corrupt each other, and because the filter cannot revise committed states, the loop runs open until divergence. Innovation gating does not repair this, because the cold-start residual is not an outlier but a systematic bias present in every early measurement: a chi-square gate rejects essentially all of them, which removes the only evidence from which $\boldsymbol{\beta}$ could be identified and leaves free-inertial drift. The window changes both halves of the failure. Each Gauss–Newton iteration relinearizes every epoch in the window, so once the coefficients begin to converge mid-window, the residual attribution and the linearization points of the early epochs are revised retroactively; and the robust kernel, unlike a gate, keeps every measurement, merely down-weighting what the current model cannot yet explain. The breadth study isolates the two effects: the window alone, with no robust kernel, already stays bounded on every counted case, so the load-bearing mechanism is the windowed relinearization, with the kernel as an accuracy refinement.

Two parameters govern the window: the length L_w and the overlap L_o. L_w sets how much data each compensation estimate sees and thus the effective adaptation time; too short and $\boldsymbol{\beta}$ is under-determined, too long and it cannot track drift. L_o sets the look-ahead and hence the commit latency: a committed state has seen L_o epochs of future data, and the estimator lags real time by L_o. In the experiments we use $L_w = 5\,\mathrm{min}$ and $L_o = 90\,\mathrm{s}$; the window-length sweep in Fig. 10 shows that shortening L_w degrades the long-line result, consistent with under-determination. The first window starts the coefficients at zero under $\mathbf{P}_0^\beta$ and lets the FOGM disturbance state absorb what the compensation cannot yet explain; subsequent windows inherit the carried covariance and converge in a few Gauss–Newton iterations (typically 3–5; Sec. V-G), which is why the per-window iteration count stays low despite the nonlinearity of the map term. How far $\mathbf{P}_0^\beta$ lets that first window travel turns out to decide whether a short-lag smoother survives the cold start at all (Section H).

C. Per-epoch incremental smoothing

Taking the lag to its lower limit turns the window into a per-epoch recursion: instead of re-solving a batch every stride, the graph is extended and re-solved once per measurement. We realize this with an incremental fixed-lag smoother built on the Bayes-tree factorization of iSAM2 [35]. At each epoch the two chain factors and the scalar-magnetometer factor of that epoch are added to the graph; iSAM2 identifies the variables whose linearization the new factors disturb and re-eliminates only that part of the Bayes tree, so the per-epoch cost does not grow with the flight. States that leave the lag are marginalized rather than dropped: their information is compressed into a Gaussian prior on the oldest retained state, the same boundary marginal the sliding window carries across its handoff. The window and the incremental smoother are therefore one estimator at two strides. The window re-optimizes a batch every $L_w - L_o$ epochs and hands the filtered boundary forward so each overlap measurement is counted once; the incremental smoother is the limit in which the stride is a single epoch, with a causal estimate available at every step from the current linearization point. Along the whole spectrum the lag alone sets how much future data corrects a committed state, hence its accuracy, latency, and per-epoch cost together.

One numerical point matters for reproduction. The position channel of the discrete process covariance $\mathbf{Q}_t$ is minute, of order $10^{-31}\,\mathrm{rad}^2$ in the horizontal-position states, because position error is driven only through velocity over a single sample; whitening the process factor by $\mathbf{Q}_t^{-1/2}$ then spans some thirty orders of magnitude. A Cholesky factorization of the information matrix squares this conditioning and fails outright, returning an indeterminate variable. Two changes remove the failure: a floor of $10^{-20}\mathbf{I}$ added to $\mathbf{Q}_t$, which injects under a millimetre of position uncertainty per step and is dynamically negligible, and a QR factorization of the square-root information matrix in place of Cholesky, which operates on the square root and so tolerates the residual spread. A larger floor or per-epoch regularizing priors also suppress the exception, but by pinning the estimate onto the propagated INS they defeat the correction; the floor-plus-QR combination keeps the estimate free while remaining numerically stable, and is what both the incremental and the batch solvers use.

D. Robust kernel and sparse solver

Map error and transient interference produce heavy-tailed residuals that a quadratic cost over-weights. We use a Huber kernel [45]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \delta, \\ \delta\big(|u| - \frac{1}{2}\delta\big), & |u| > \delta, \end{cases} \tag{11}$$

minimized by iteratively reweighted least squares (IRLS): each Gauss–Newton iteration reweights measurement factor t by

$$w_t = \min\big(1,\ \delta/|u_t|\big), \qquad u_t = r_t/\sqrt{R}, \tag{12}$$

so gross outliers are down-weighted toward an absolute-value penalty while inliers keep the efficient quadratic cost. In the implementation the reweighting is applied from the second Gauss–Newton iteration onward (the first iteration is plain least squares) with w_t clamped to $[10^{-6}, 1]$, and $\delta = 1.345$ gives the standard 95% Gaussian efficiency. Alternatives such as switchable constraints [46] and dynamic covariance scaling [47] fit the same graph interface and could replace (11) unchanged.

Each Gauss–Newton iteration solves a sparse linear system whose Jacobian has the block-bidiagonal structure of the two chains in Fig. 4 (in the static-$\boldsymbol{\beta}$ limit the coefficient blocks collapse to one shared column). We factor it by column-pivoted QR on the square-root information matrix (square-root SAM [33], [28]), which is numerically stable and exploits the chain sparsity. In the Gaussian case an iterated RTS pass [43] returns the identical estimate:

Lemma 1:
Under Assumption 1, with $\rho(u) = \frac{1}{2}u^2$ and $\boldsymbol{\beta}$ held fixed, the minimizer of (8) equals the Rauch–Tung–Striebel (RTS) smoother estimate over the window.

Proof:
With ρ quadratic and $\boldsymbol{\beta}$ fixed, (8) is the negative log-density of a linear–Gaussian chain, so its minimizer is the posterior mean, which the forward Kalman and backward RTS passes compute [43]. ■

The lemma serves two ends. It licenses the fast iterated-RTS solve on the navigation chain, which we check against the QR path to within a metre in Section V-A; and it fixes scope, since the paper's gain comes from precisely what the lemma's hypotheses exclude: the jointly estimated, time-varying $\boldsymbol{\beta}$, the robust kernel (11), and the fixed-lag window.

IV. Separability of Position and Compensation

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let $\mathbf{G}$ have rows $\mathbf{g}_t^\top$ (the map-gradient/position sensitivity) and let $\boldsymbol{\Psi}$ have rows $\mathbf{A}_t^\top$ (the TL regressor). To isolate the position–compensation coupling, restrict (3) to the $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ subspace, giving the window model $\mathbf{z} = \mathbf{G}\,\delta\mathbf{p} + \boldsymbol{\Psi}\,\delta\boldsymbol{\beta} + \boldsymbol{\eta}$ (Fig. 6). This model freezes $\delta\mathbf{p}$ and $\delta\boldsymbol{\beta}$ over the window: it is a per-window surrogate for the time-varying system (2), (9) that isolates the position–compensation coupling. It holds the coefficients fixed and sets aside the slow magnetic-bias state S_t and the INS dynamics that a full observability Gramian would mix in, so it speaks to how position and compensation are told apart, not to observability of the complete time-varying state.

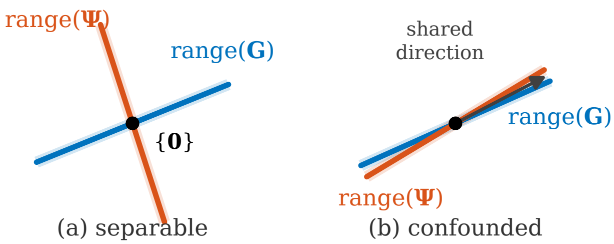

Fig. 6. Observability geometry (Proposition 1): position and compensation are separable when range($\mathbf{G}$) and range($\mathbf{\Psi}$) meet only at the origin (left) and confounded when they share a direction (right).

Proposition 1:
A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment, hence unobservable from map matching over the window, if and only if $\mathbf{G}\,\delta\mathbf{p} \in \operatorname{range}(\mathbf{\Psi})$. Consequently the joint state $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ is observable if and only if the stacked matrix $[\,\mathbf{G} \quad \mathbf{\Psi}\,]$ has full column rank; equivalently, if and only if all three of the following hold: (i) $\ker \mathbf{G} = \{\mathbf{0}\}$ (the map gradient excites every position direction over the window); (ii) $\ker \mathbf{\Psi} = \{\mathbf{0}\}$ (the compensation basis has full column rank); and (iii) $\operatorname{range}(\mathbf{G}) \cap \operatorname{range}(\mathbf{\Psi}) = \{\mathbf{0}\}$ (no shared direction). Condition (iii) alone governs the position–compensation confounding: when it fails, a position shift with $\mathbf{G}\,\delta\mathbf{p} \neq \mathbf{0}$ is reproduced exactly by a compensation change.

Proof sketch:
Two joint states are indistinguishable exactly when their difference lies in $\ker[\,\mathbf{G} \quad \mathbf{\Psi}\,]$, so observability is full column rank of the stacked matrix; sorting the kernel elements by whether $\mathbf{G}\,\delta\mathbf{p}$ vanishes yields conditions (i)–(iii). The full proof is in Appendix C. ■

Condition (iii) is the operative one in practice. Conditions (i) and (ii) are routine to check; the confounding of position with compensation, by contrast, depends on the joint geometry of the flight path and the local map gradient, and it can fail only through coincidence of the flown attitude history with that gradient, plausible, for instance, when turns always occur at the same map features. One expects (iii) to hold for trajectories and maps in general position, but the condition is non-constructive: it yields no reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection (in-sample basis R^2, out-of-sample prediction, gradient-alignment indices) did not generalize across estimators or lines. We report that as a negative result, not a predictive test.

Remark 1:
Condition (ii) fails, or very nearly fails, for the classical TL basis itself: with unit direction cosines the three induced diagonal columns sum to $B_t(u^2+v^2+w^2) = B_t$, which is collinear with the constant-bias column to the degree that the total field is constant over a window (B_t varies by well under a percent of its $5 \times 10^4\,\mathrm{nT}$ magnitude), the long-known collinearity of TL regression [12], [18]. This is, to that approximation, a gauge freedom of the compensation rather than a navigation defect: a perturbation $\delta\boldsymbol{\beta}$ in the near-null subspace of $\mathbf{\Psi}$ moves the predicted interference $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$, and hence the position estimate, only negligibly. The prior $\mathbf{P}_0$ and the random-walk factors select one representative of each near-gauge orbit. Separability of position from compensation is governed by (iii) alone and is unaffected.

Remark 2:
The condition is geometric: it depends on the ranges of $\mathbf{G}$ and $\mathbf{\Psi}$ over the window, hence on the flight path and the local map gradient. It explains the dominant failure mode seen in practice: over a locally flat map $\mathbf{G} \approx \mathbf{0}$ and condition (i) fails, so position is unobservable regardless of the compensation, a map-information floor that bounds every estimator, which we observe on the shortest line in Section V.

The cold-start transient observed in real time makes this confounding visible. A per-epoch smoother with a short lag (Section C), run from a cold start on line 1007.06 Mag 5, holds a horizontal DRMS near $82\,\mathrm{m}$ through the first several minutes before locking on (Fig. 7). Seeding the compensation with a warm Tolles–Lawson prior, tightening its initial standard deviation from 1.0 to 0.1, removes only about $40\,\%$ of this transient, to roughly $50\,\mathrm{m}$; the remainder is not a compensation error but a map-observability one, precisely conditions (i) and (iii) of Proposition 1 early in the flight. Two observations separate the causes. First, the same cold start with a long lag shows no transient at all: given enough future gradient, the smoother resolves the position–compensation ambiguity that the short lag cannot. Second, the cold and warm runs lock on at the same instant, when the aircraft crosses a strong map feature about four and a half minutes in, rather than after a fixed compensation-learning time. Both point to the map gradient, not the compensation prior, as what ultimately breaks the confounding: where $\operatorname{range}(\mathbf{G})$ is rich the two unknowns separate, and a warm prior only shortens the wait; Fig. 7 shows the decomposition on the first six minutes of the segment.

That decomposition holds for a magnetometer whose uncompensated field the compensation prior can represent. It does not hold for the cabin magnetometer with the larger platform field, where the prior itself is the binding constraint and neither a longer lag nor a richer map gradient recovers the estimate; Section H separates that case and reports the settings under which it is bounded. The two statements are about different regimes and we keep them apart: in the first regime the compensation can reach the interference and is merely slow; in the second it cannot reach it at all.

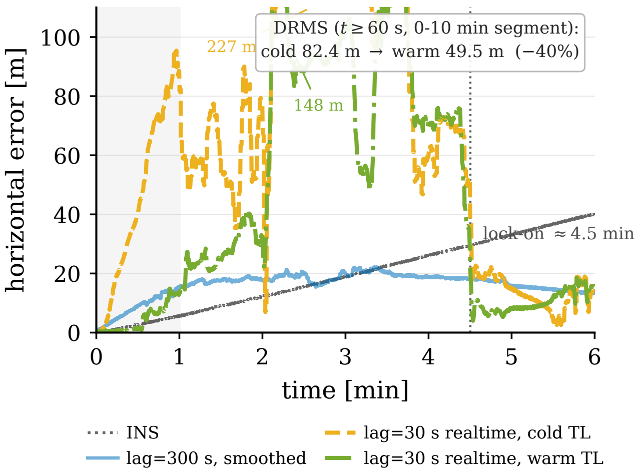

Fig. 7. First six minutes of the cold-start transient at a 30 s lag: a warm compensation prior cuts the penalty by 40 %, but both runs lock on only at the strong map feature near four and a half minutes.

TABLE I
Estimator settings for all cold-start results; the same $\mathbf{P}_0$, $\mathbf{Q}$, R, and TL settings drive both the EKF and the smoother.

Group	Parameter	Value
Measurement	noise std. $\sqrt{R}$	$12\,\mathrm{nT}$
FOGM S	(σ_S, τ)	$3\,\mathrm{nT}$, $180\,\mathrm{s}$
Initial $\mathbf{P}_0$	pos. / alt. std.	0.1 / $1.0\,\mathrm{m}$
	velocity std. (per axis)	$1.0\,\mathrm{m\,s^{-1}}$
TL prior / walk	$\mathbf{P}_0^{\beta}$; $\mathbf{Q}^{\beta}$	$\mathbf{I}(\sigma_\beta)^2$; $\mathbf{I}(1.0)^2\Delta t$
	σ_β (window, batch, Sec. IV)	1.0
	σ_β (per-epoch, Sec. H)	100
TL basis	terms (App. B)	P+I+E+bias (19)
Window	L_w / L_o / Δt [s]	300 / 90 / 0.1
Robust	Huber δ	1.345
	IRLS	2nd iter. on, $w \in [10^{-6}, 1]$
Gauss–Newton	max. iter. / tol.	5 / 10^{-4}

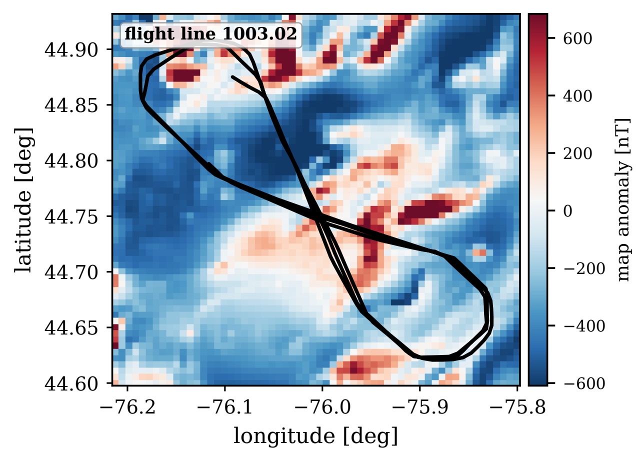

Fig. 8. Line 1003.02 over the Eastern anomaly map.

V. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [14]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically-pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines accordingly. Flt1007 is the free-flight mission, in which the aircraft flies freely within the mapped Renfrew ($\approx 400\,\mathrm{m}$) and Eastern/Perth ($\approx 800\,\mathrm{m}$) areas, so its lines are the closest to a navigation task and are what prior work [14], [4], [19] reports; 1007.06 is our primary line for that reason. Flt1003 is a crustal-field measurement flight (survey traverses at 400 and $800\,\mathrm{m}$); its lines 1003.02 and 1003.08 are straighter than the free-flight legs. Flt1006 is a calibration-maneuver flight, so we set its single short line aside, and the map-building flights (Flt1004–1005) are not used. Two honest points remain. First, none of these is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry more attitude variation, which stresses compensation but also aids TL observability by exciting the regressor, so performance on a pure operational cruise is not directly established here and is future work. Second, the lines span two altitude regimes: four at the $\approx 400\,\mathrm{m}$ survey-drape altitude and 1007.02 at $\approx 800\,\mathrm{m}$, where upward continuation attenuates the anomaly. Map matching uses the Eastern ("E") and Renfrew ("R") high-resolution anomaly grids, upward-continued to flight altitude with the IGRF core field restored. The metric is horizontal distance-RMS (DRMS) of position error after a $10\,\mathrm{min}$ warm-up. Cold start means the TL coefficients are initialized to zero with no calibration flight; the navigation solution is accurately initialized (position std $0.1\,\mathrm{m}$, Table I), so the term denotes a compensation cold start, not an unknown-position navigation start. For every comparison the EKF and FGO estimators use identical dynamics, map interpolation, and noise settings (Table I); the window smoother additionally applies the Huber kernel (11), whose isolated contribution is reported separately in Table II ($33.0 \to 30.7\,\mathrm{m}$ on Mag 4), so the reader can attribute the remaining margin to the estimator structure. Unless noted, the window smoother uses $L_w = 5\,\mathrm{min}$ and $L_o = 90\,\mathrm{s}$; L_w was selected on line 1007.06 (Fig. 10) and then held fixed for every other line, so the breadth results of Table III are not per-line tuned. All numbers are produced by scripted runs under a fixed software environment and are available from the authors on request.

For geographic context, Fig. 8 shows line 1003.02 over the Eastern anomaly field. As a controlled check on the compensated stinger, before the cold-start problem, the batch factor graph tightens the horizontal DRMS from the EKF's $31.5\,\mathrm{m}$ to $14.0\,\mathrm{m}$, against $114.4\,\mathrm{m}$ free-inertial (Fig. 9); its iterated-RTS and sparse-GN/QR solvers agree to within $1\,\mathrm{m}$, an empirical check of Lemma 1.

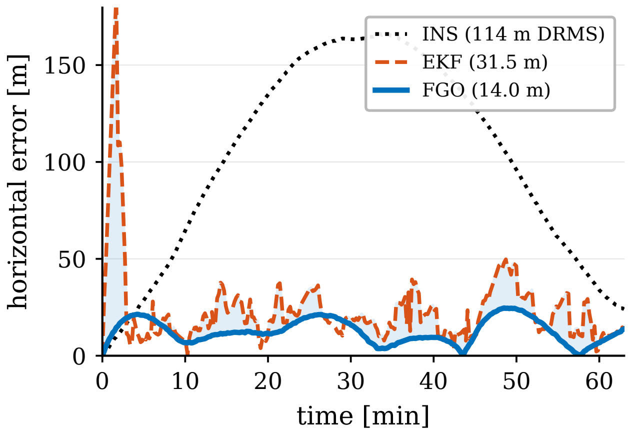

Fig. 9. Horizontal position error on line 1003.02: batch FGO, EKF, and free inertial (DRMS in the legend).

B. Baselines

The strong causal competitor is the online EKF+TL+NN lineage that augments the Pinson error state with the TL coefficients β and the weights of a small feed-forward network, all propagated as filter states and updated online [15], [19]. We run its reference implementation [15], tuned by us for a stable cold start, and do not claim to reproduce the extended filter of [19]. Cold start forces one substantive change: with no calibration flight to warm-start from, we initialize the network output normalization data-drivenly from the first five minutes of onboard interference; a naive port instead drives the network bias to $\sim 5 \times 10^4\,\mathrm{nT}$ and the filter to `NaN`. We keep the uncompensated scalar out of the network inputs, as Proposition 1(iii) requires: a position-dependent input carries sensitivity $\mathbf{g}_t$, so the network absorbs the map signal and position drifts while the residual stays small (the scalar-augmented variant [19] escapes this only by freezing the network after its transient). The tuned filter uses the permanent TL group with an eight-unit hidden layer, a FOGM $(\sigma, \tau) = (3\,\mathrm{nT}, 180\,\mathrm{s})$, and measurement variance $12^2\,\mathrm{nT}^2$, reaching $48.8/18.3\,\mathrm{m}$ on Mag 4/5 (Table II), consistent with published cold-start operating points [15], [19]. We also compare a weak online-TL EKF and a marginalized particle filter carrying the TL coefficients; all are our own runs under one protocol, except the particle filter, reported under its most favorable per-case configuration.

C. Cold-start line vs. online EKF+TL+NN

The central comparison is the cold start on uncompensated cabin magnetometers, where the interference is present and must be learned online. Table II and Fig. 11 report the primary evaluation line 1007.06 ($87\,\mathrm{min}$, full length) on Mag 4 and Mag 5. A static-TL batch forces a single coefficient vector over the whole line; it underfits the time-varying interference and degrades to $123.7/68.1\,\mathrm{m}$. Allowing the compensation to adapt with a

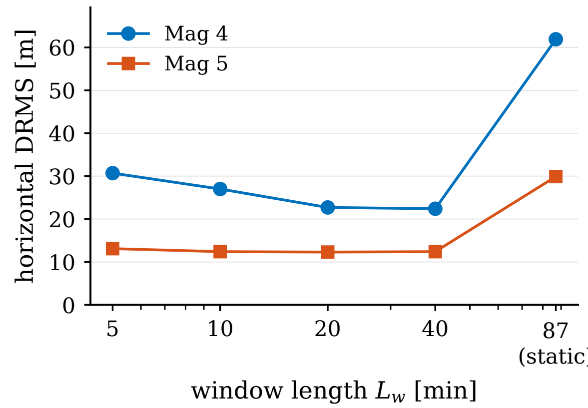

Fig. 10. Window length L_w versus horizontal DRMS on line 1007.06 (cold start).

TABLE II
Cold start on line 1007.06 (Flt1007), uncompensated cabin magnetometers; DRMS [m]. INS $318\,\mathrm{m}$; compensated-Mag 1 EKF $\approx 20\,\mathrm{m}$.

Method (no NN unless noted)	Mag 4	Mag 5
FGO-online, static batch TL	123.7	68.1
FGO-online, window $5\,\mathrm{min}$	33.0	13.5
FGO-online, window $5\,\mathrm{min}$ + Huber	**30.7**	**13.1**
EKF+TL+NN, online [15], [19]	48.8	18.3

$5\,\mathrm{min}$ window recovers it to $33.0/13.5\,\mathrm{m}$, and the Huber kernel further trims transient outliers to $30.7/13.1\,\mathrm{m}$. Our online EKF+TL+NN baseline reaches $48.8/18.3\,\mathrm{m}$. The window smoother beats the NN-augmented filter on both sensors, $18.1\,\mathrm{m}$ on Mag 4 and $5.2\,\mathrm{m}$ on Mag 5, without any neural network, using only the linear TL basis carried as graph variables. All compared numbers are our own runs under one protocol; 1007.06 is the line on which the baseline tuning is anchored, and the same baseline carries to the other lines in the breadth study (Table III). We attribute the margin to the window look-ahead: the coefficients are identified from data on both sides of each committed epoch, whereas the EKF must commit with past data only.

The window length itself trades under-determination against staleness. Fig. 10 plots DRMS across window lengths on this line: the full-line static batch cannot track the coefficient drift, windows that are too short under-determine them, and the $5\,\mathrm{min}$ window and longer stay bounded, with $5\,\mathrm{min}$ balancing adaptation against under-determination. The tradeoff mirrors the expressiveness discussion of Section IV and makes L_w a physically interpretable knob rather than a free hyperparameter; a denser sweep across more lines, where the balance can shift toward longer windows, is left to future work.

These cold-start numbers also settle a question the formulation raises: with no neural network, how does the estimator cope with the large fraction of the cabin interference that the linear TL basis cannot span. The mechanism is the one built in Section III-A, where the

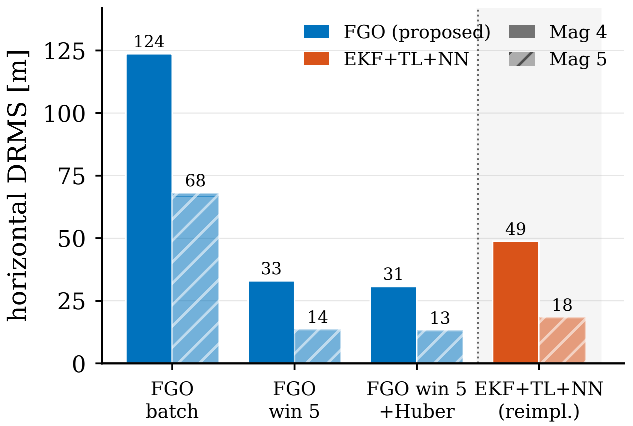

Fig. 11. Cold start on line 1007.06: NN-free fixed-lag window (left) versus the online EKF+TL+NN baseline (right).

time-varying coefficients, the FOGM disturbance state, and the robust kernel together absorb that remainder as a structured robust regression. The trade it makes is visible in the numbers. The network buys a lower compensation residual, while the window, the disturbance state, and the robust kernel together buy navigation robustness at a higher residual. This is why the uncompensated Mag 4 DRMS of tens to hundreds of metres sits well above survey-grade compensation and still bounds the inertial drift. A learned residual could be added later as an extra factor on the same measurement, for example a Gaussian-process or small-network term, which would place the network's expressiveness inside the graph rather than inside a filter; we leave that to future work.

D. Breadth across lines, flights, and maps

Table III runs the same cold-start TL model on four long lines spanning two flights and two maps, free-flight navigation legs (Flt1007) and survey traverses (Flt1003), grouped by purpose, against two linearized baselines: the plain online-TL EKF and the online EKF+TL+NN of Section B (the strong causal method that a fair comparison demands). A marginalized (Rao-Blackwellized) particle filter carrying the TL coefficients in its conditionally-linear-Gaussian block (MPF+TL), a genuine non-Gaussian Bayesian estimator rather than a linearization, is examined separately below.

The baselines behave very differently, and the contrast is the point. The plain online-TL EKF is fragile: on the noisy Mag 4 it diverges from the cold start to tens of kilometres on two lines and runs off the map on a third. The neural network is what fixes this: the EKF+TL+NN stays finite on all eight cases, confirming that its residual network is what buys cold-start robustness for the causal filter. The window smoother matches that robustness without any neural network (it is bounded on all eight) and improves on accuracy: across the four full-length lines ($63\,\mathrm{min}$ to $87\,\mathrm{min}$, eight mag-cases) it is the best of the four estimators on all eight, beating the NN-augmented filter by $5\,\mathrm{m}$ to $90\,\mathrm{m}$ (Fig. 12); the particle filter, reported separately, diverges on these cases.

The marginalized particle filter is reported separately because it resists a single-number summary. Implemented as the reference RBPF (bootstrap proposal, systematic resampling, and the same conditionally-linear TL block) and given its most favorable per-case configuration, with the measurement noise matched to the residual, resampling at half the effective sample size, and regularization roughening, it is erratic at cold start: bounded near $40\,\mathrm{m}$ only on the cleanest Mag 5 legs, degrading to hundreds of metres or kilometres, with a strong run-to-run (seed) dependence, elsewhere. With the toolbox default settings it diverges outright. On no counted case does it come within a factor of two of the window, and its position covariance is severely over-confident (the consistency study below). The sensitivity is itself the finding: a sequential particle representation needs the case-by-case tuning to stay bounded at cold start that the batch window does not, so a non-Gaussian representation alone does not substitute for the re-smoothing.

We also ran the short calibration-maneuver line 1006.08 (Flt1006) but set it aside rather than tabulate it: it is only $14\,\mathrm{min}$ long, so the $10\,\mathrm{min}$ warm-up leaves a $4\,\mathrm{min}$ window over which none of the DRMS values is reliable, and its low map-information floor (the Remark in Section IV) makes every method weak. It reproduces the same causal-EKF divergence, but we draw no accuracy conclusion from it: in particular the NN's $108.3\,\mathrm{m}$ on its Mag 5, against $117\,\mathrm{m}$ to $122\,\mathrm{m}$ for the TL-only EKF and window, is not evidence of an NN accuracy advantage on a window this short.

Two effects in the reliable cases are worth naming. First, the EKF+TL+NN follows the reproduced design in feeding the network only the permanent TL group, whereas the online-TL EKF and the window smoother both carry the full permanent-induced-eddy-bias basis; on the well-conditioned Mag 5 those linear higher-order terms matter, so the NN filter can fall behind even the plain EKF (1003.02: 29.5 vs $28.1\,\mathrm{m}$) when its network has not yet learned what the dropped linear terms would have supplied directly. Second, and relatedly, the network's extra online-learned states add estimation variance where the linear basis already suffices, a cost the NN-free window avoids. Both effects favor a linear compensation carried in a smoother over a learned one carried in a filter, on this benchmark. One caveat on provenance: the EKF+TL+NN is the lineage's reference implementation under our tuning, matching the reported 1007.06 operating points to within a few metres, but the original's per-line numbers on the other lines are unpublished, so the comparison is against a faithful, not the authors', filter. Its cold-start trajectory is also numerically sensitive, so its per-line DRMS varies between runs; the values here are those of a single fixed run and do not affect the best-of-three ordering.

Two conclusions follow, and they are worth separating because a single-line comparison conflates them. The first

TABLE III

Breadth: cold-start cabin magnetometers, DRMS [m]. Purpose FF = free-flight (Flt1007), SV = survey (Flt1003); "div." $> 10\,\mathrm{km}$, "err." = off map; bold = best of three. The marginalized particle filter (MPF+TL) is treated separately in the text: even at its most favorable per-case tuning it is erratic and never competitive with the window. Calibration line 1006.08 set aside (see text).

Line	Purp.	Map	alt [km]	Mag	EKF online	EKF+ TL+NN	FGO win
1007.06	FF	R	0.4	4	46.7	48.8	**30.7**
1007.06	FF	R	0.4	5	17.8	18.3	**13.1**
1007.02	FF	E	0.8	4	div.	130.0	**39.9**
1007.02	FF	E	0.8	5	31.6	30.4	**15.4**
1003.02	SV	E	0.4	4	div.	101.0	**43.1**
1003.02	SV	E	0.4	5	28.1	29.5	**21.1**
1003.08	SV	R	0.4	4	err.	46.6	**28.0**
1003.08	SV	R	0.4	5	21.1	20.7	**13.1**
FGO best (all counted lines)							**8 / 8**

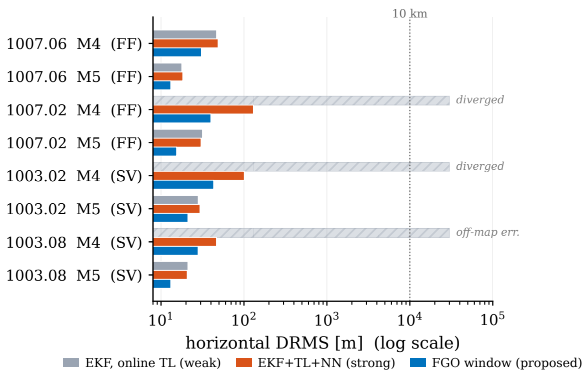

Fig. 12. Breadth study (Table III), log DRMS axis: online-TL EKF (gray), EKF+TL+NN (red), and the NN-free FGO window (blue). The MPF+TL particle filter is erratic at cold start (see text) and is omitted.

is about robustness. Keeping a causal filter bounded at cold start is not unique to the window, since the neural network does it too, so the blanket claim that causal filters diverge is false for the state-of-the-art filter. It is, however, true of every non-NN cold-start estimator we tried: the plain online-TL EKF diverges, and the marginalized particle filter, though a full Bayesian estimator, is only erratically bounded and never competitive (preceding paragraph). Reliable cold-start compensation without a network requires the batch/window re-smoothing, not merely a non-Gaussian representation. An ablation within our own estimator locates the mechanism more precisely: running the same window with the robust kernel disabled still stays bounded on all eight counted cases (e.g. $60.3\,\mathrm{m}$ on 1003.02/Mag 4 where the causal filter reads $41\,626\,\mathrm{m}$, refined to $43.1\,\mathrm{m}$ by Huber), so the robustness comes from the windowed relinearization of Section III-B, and the kernel buys accuracy, not survival. The recorded platform currents let us check the absorption account of Section III-A directly on line 1003.02: the fourteen onboard current channels linearly explain $28\,\%$ to $39\,\%$ of the raw cabin interference (strobe lights, cabin fuel pump, and INS heater lead), but only $2\,\%$ to $6\,\%$ of the post-fit windowed-FGO residual, so the current-aligned component is almost entirely absorbed by the time-varying $\boldsymbol{\beta}_t$ and S_t; and the Huber-down-weighted epochs concentrate 2.6–$3.6\times$ on current-switching activity, with strobe switching alone correlating at 0.55 with the residual magnitude on Mag 5, confirming that the kernel attenuates precisely the transients the linear states cannot track. These are instantaneous linear regressions, hence lower bounds on the current contribution. What the window adds is that it reaches the same bounded regime without the network, its online-training machinery, or the verification burden that comes with it. The second is about accuracy, where the window is the best of the four on all eight full-length cases; under a paired sign test on the per-case DRMS the one-sided probability of this count arising by chance is $2^{-8} \approx 0.004$. This margin is real, but it should be read with care, because our EKF+TL+NN is a tuned reference implementation and part of its deficit is that its online-learned states add variance on the well-conditioned sensor, where the linear TL basis already suffices. The defensible reading is that a linear estimator with no network, given future context through the window, is enough both to survive the cold start and to match or beat an NN-augmented causal filter on this benchmark. Table IV places the estimator families along the axes that matter for cold-start compensation.

TABLE IV

Estimator families for cold-start map-matching compensation. ✓ = yes, ✗ = no, (∼) = limited.

Estimator	Future data	Comp. state	Robust meas.	Bounded latency
Causal EKF [4]	✗	(∼)	✗	✓
Particle / grid filter [8]	✗	✗	✓	✓
Online EKF+TL+NN [15], [19]	✗	✓	(∼)	✓
Batch FGO	✓	✓	✓	✗
Proposed (fixed-lag FGO)	✓	✓	✓	✓

The separability condition of Section IV accounts for the pattern. The map-information floor (Remark 2) is why the short calibration line 1006.08 is hard for every method, since there is little independent map gradient to exploit over its brief window, and it is the reason we set that line aside. The condition is geometric and predictive only through these qualitative rules; a scalar screening statistic we tried did not generalize, which we report as a negative result.

E. Consistency and statistical significance

Each real line provides a single physical realization, so its DRMS is a point estimate and its measurement noise cannot be resampled. To assess statistical significance and, more importantly, whether the reported covariance is credible, we run a Monte-Carlo study in simulation, where repeated realizations are valid. We fix one simulated trajectory over the high-resolution Eastern anomaly map and, for each of 30 seeds, draw an independent INS-error realization and an independent clean scalar-measurement realization, then run three map-matching estimators on the same draw: the causal EKF, the marginalized (Rao-Blackwellized) particle filter already in the toolbox, and the batch factor graph. The clean compensated measurement isolates the navigation covariance from the cold-start compensation transient.

Averaged over the 30 seeds, after a $30\,\mathrm{s}$ warm-up, the horizontal DRMS is $1.4\,\mathrm{m}$ for the factor graph, $4.1\,\mathrm{m}$ for the EKF, and $4.9\,\mathrm{m}$ for the particle filter, with $95\,\%$ confidence half-widths of 0.2, 0.4, and $0.8\,\mathrm{m}$. Even on this clean sensor the factor graph separates cleanly from both recursive baselines (its interval does not overlap theirs), while the EKF and the particle filter are statistically indistinguishable in accuracy. The larger cold-start separations are in Sections V-C and V-D; the purpose of this study is consistency.

For consistency we use the per-axis two-degree-of-freedom position NEES, $(e_N/\sigma_N)^2 + (e_E/\sigma_E)^2$, whose expectation under a consistent covariance is 2. Averaged over all seeds and epochs the ANEES is 1.57 for the factor graph and 1.72 for the EKF, both below the ideal 2: both report conservative uncertainty that bounds the true error with margin. The toolbox particle filter is the opposite: its ANEES is 59.8, more than an order of magnitude above 2, the particle-depletion signature in which the surviving particles after resampling on a high-resolution map cluster far more tightly than the true error and the empirical covariance collapses. This overconfidence is a configuration artifact, not a property of the estimator class: adding regularization roughening restores consistency (ANEES 1.8 at $1\,\mathrm{m}$ roughening) at a small accuracy cost. Even so, the factor graph dominates both recursive baselines, more accurate (1.4 versus $4.1\,\mathrm{m}$ to $4.6\,\mathrm{m}$) and consistent at once. Fig. 13 shows the EKF and factor-graph NEES staying inside the $95\,\%$ χ^2_2 band throughout a representative run while the toolbox particle filter runs far above it, and Fig. 14 shows the North and East position error contained within the factor-graph $\pm 2\sigma$ envelope. Two caveats bound the reading. The NEES uses the per-axis (diagonal) covariance, since the cross term is not exposed by the evaluated filter output, and the study is a matched-model simulation, so it certifies internal consistency rather than robustness to model mismatch. The conservative bias of the linearized estimators indicates that their process noise could be tightened, which is the safe direction in which to err.

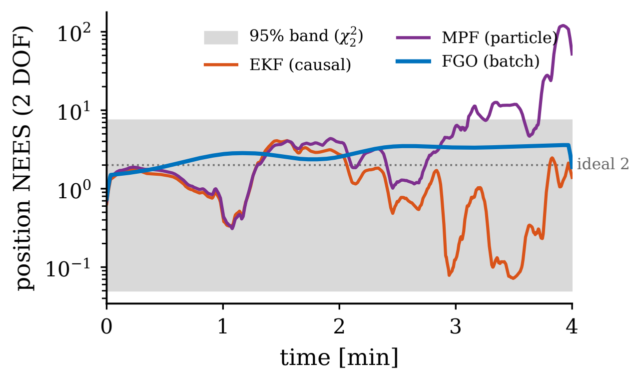

Fig. 13. Monte-Carlo position NEES versus time (EKF, marginalized particle filter, and batch FGO) with the $95\,\%$ χ^2_2 band (ideal 2, dotted). NEES divides each estimator's error by its own reported covariance, so a lower curve means a more conservative filter, not a more accurate one; the EKF's larger errors over its larger causal covariance can plot below the more accurate smoother. Accuracy is compared by DRMS in the text.

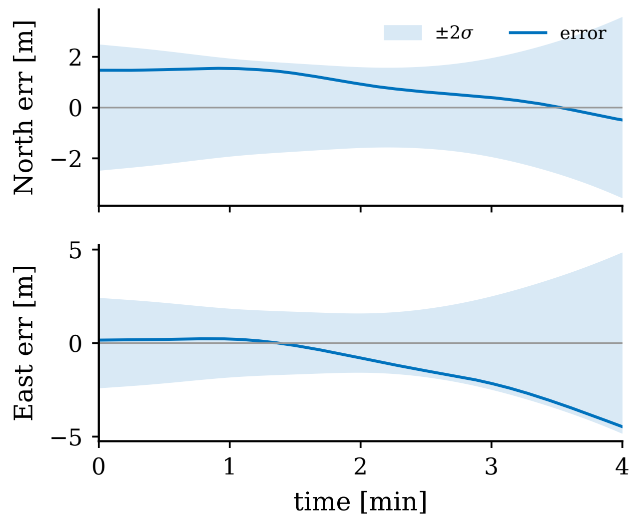

Fig. 14. North and East position error within the estimator $\pm 2\sigma$ bands (representative simulated run of the batch FGO; the envelope is the smoother's own reported uncertainty, so this checks credibility, not accuracy). A calibrated Gaussian $\pm 2\sigma$ band would be exceeded on $\approx$5% of epochs per axis; here the error approaches but never exits the band, the mildly conservative behaviour consistent with the ANEES below 2.

F. Cross-implementation validation

Because every number above comes from our own code, a fair question is whether the advantage belongs to the estimator or to one implementation. To separate the two, we reran the primary line 1007.06 through a second, independent estimator: a Python factor graph built on the GTSAM library [29], with its own Levenberg–Marquardt and iSAM2 solvers, sharing no code with the in-house Julia smoother, whose back end is an iterated Rauch–Tung–Striebel and Gauss–Newton pass. Both solve the same factor graph of Section III from the same inputs. We first checked the port itself: the state-transition slices $\mathbf{\Phi}_t$, the regressor rows $\mathbf{A}_t$, the scalar measurements, and the INS mechanization match to the bit across the two languages, and the interpolated anomaly-map grid agrees to a $0.9\,\mathrm{nT}$ standard deviation, so the two estimators see the same problem.

On the full $87\,\mathrm{min}$ cold-start line, the two land within a metre of each other: the GTSAM window returns $12.4\,\mathrm{m}$ horizontal DRMS against the Julia smoother's $13.1\,\mathrm{m}$ under the marginalized handoff both use (Table III; the legacy smoothed handoff gives $13.8\,\mathrm{m}$), and on the first ten-minute segment 17.5 against $21.4\,\mathrm{m}$, the GTSAM side being slightly the better of the two in both cases; the free-inertial drift matches at 317.5 against $318\,\mathrm{m}$, a further check that the two share the same trajectory and map. The residual differences trace to modeling choices left deliberately independent: Levenberg–Marquardt versus plain Gauss–Newton, bicubic versus bilinear map interpolation, and a slightly different window handoff. The comparison covers one line and each side is a single run, so this is a statement about implementation independence rather than a statistical one; its value is that the bounded cold-start result survives a complete change of language, library, and solver. Fig. 15 overlays the GTSAM error trace on the segment.

On the cabin magnetometer with the larger platform field the agreement is conditional, and what it is conditional on is instructive. Run on Mag 4 of the same line from a zero compensation, the GTSAM window returns $5430\,\mathrm{m}$ against the Julia smoother's $30.7\,\mathrm{m}$, with identical inputs and the identical prior; on 1003.02 Mag 4 it returns $36.6\,\mathrm{m}$ against $43.1\,\mathrm{m}$ and there is no discrepancy at all. The full-line GTSAM batch on 1007.06 Mag 4 returns $18.6\,\mathrm{m}$, so neither the graph nor its cost function is at fault. Seeding the first window's compensation with the ridge fit of Section I, which changes nothing else, brings the GTSAM window on that case to $30.8\,\mathrm{m}$ against the Julia smoother's $30.7\,\mathrm{m}$. The two implementations therefore agree on both magnetometers once the compensation cold start is handled, and the earlier disagreement was where the first window's compensation started, not the language, the library, or the solver. That the two behave differently from the same start is consistent with the sensitivity of Section H: the Julia smoother stops after at most five Gauss–Newton passes per window while the GTSAM window runs Levenberg–Marquardt to convergence, so they sample that sensitivity differently. We report that as the difference we observe, not as a mechanism we have isolated.

G. Real-time incremental smoothing

Each window is a sparse nonlinear least-squares problem of fixed size $O\big(L_w(n+m)\big)$; the block-bidiagonal Jacobian of the two chains (Fig. 4) is factored in $O(L_w)$ by the square-root solver [33], and the number of Gauss–Newton iterations is small (typically 3–5) because the problem is mildly nonlinear away from observability collapse. The per-window cost is therefore constant in the line length and the overall complexity is linear in the number of epochs, matching a filter's asymptotic cost while retaining the look-ahead. Memory is bounded by the window, not the flight. The one-time overhead relative to an EKF is the window factorization and the IRLS reweighting, which in our unoptimized research implementation runs comfortably faster than real time on the SGL sampling rate.

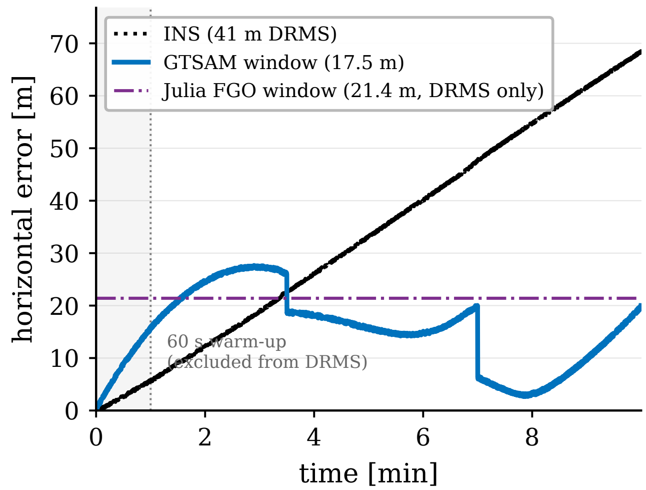

Fig. 15. Horizontal error of the independent GTSAM sliding window on the first ten-minute segment of line 1007.06, with the Julia smoother's segment DRMS shown as a reference level (no Julia time series was exported for this run).

H. The compensation prior at cold start

Taken to its per-epoch limit, the smoother exposes a modeling choice that the full-line batch tolerates. Run from a zero compensation on the cabin magnetometer with the larger platform field (Mag 4), the incremental smoother at a $300\,\mathrm{s}$ lag leaves the map on four of the five lines, three of the four counted ones, and ends between 6.6 and $27\,\mathrm{km}$ from truth, while the same smoother on Mag 5 of the same flights navigates normally (Table V, first column). The asymmetry is the clue, and it is one of magnitude. Over the first $300\,\mathrm{s}$, the residual against the map at the INS position is $1729\,\mathrm{nT}$ to $2081\,\mathrm{nT}$ root-mean-square on Mag 4 and $22\,\mathrm{nT}$ to $314\,\mathrm{nT}$ on Mag 5, and the compensation coefficients that explain the Mag 4 field have magnitudes of 552 to 722 in the units of Appendix B.

Against that, the priors the model offers for anything other than position are small. The compensation prior is $\mathbf{P}_0^{\beta} = \mathbf{I}(\sigma_\beta)^2$ with $\sigma_\beta = 1$, one scalar shared by all 19 coefficients, and the FOGM disturbance prior is $3\,\mathrm{nT}$. A cold start on Mag 4 therefore presents the graph with roughly $2000\,\mathrm{nT}$ while both nuisance channels are stiff, and the degree of freedom that is not stiff is position, through the map gradient. The single shared σ_β is the weaker part of that: by Appendix B the regressor columns range over four orders of magnitude, from direction cosines of order one to induced and eddy terms carrying the field magnitude, so no single coefficient scale can be right for all of them.

Table VI measures the sensitivity. Widening σ_β with the prior mean left at zero and nothing else changed removes the divergence on every Mag 4 line: $\sigma_\beta = 10$ already suffices and 100 to 1000 changes the result by up to $14\,\mathrm{m}$. We recommend $\sigma_\beta = 100$ and use it for the rest of this subsection: the Mag 4 lines settle at 19.6, 25.0, 99.1, 28.4 and $24.2\,\mathrm{m}$ and the Mag 5 lines at 12.0, 10.5, 101.0, 15.8 and 11.5 m. A per-column prior, one that assigns each basis column a standard deviation of 100 nT of measurement contribution instead of a common coefficient scale, is the dimensionally consistent version of the same change and performs the same to within the line-to-line spread (Table VI). It does not, however, keep the coefficients smaller: after warm-up the median of $\|\boldsymbol{\beta}\|_\infty$ is 787 to 1167 across the Mag 4 lines at $\sigma_\beta = 100$ and 2485 to 5995 under the per-column prior, against the 552 to 722 that a ridge fit of the same data needs. Any prior loose enough to fix the cold start also lets the compensation basis absorb model error beyond the platform field, and we report that as a cost of the setting rather than a property we have controlled. Widening the FOGM disturbance prior and walk instead, leaving the compensation prior nominal, recovers three of the five lines at a factor of 10 and four at a factor of 100, leaving 1006.08 diverged in both cases (Table VI), so the effect is not specific to the compensation block, although the compensation basis is where the interference belongs.

We do not claim more than the sensitivity we measured. In particular the mechanism is not simply that the coefficients cannot reach the platform field: the window solves of Section I reach $\|\boldsymbol{\beta}\|_\infty$ of 463 to 795 across lines at $W \geq 100$ s even at $\sigma_\beta = 1$, and the prior term at that magnitude is outweighed by the measurement term by more than two orders of magnitude. What the tight, dimensionally inconsistent prior does is stiffen the first linearizations in $\boldsymbol{\beta}$ relative to position, and the per-epoch smoother commits and marginalizes states while that is true.

Seven alternatives can be dismissed by measurement, all at the nominal prior on line 1007.06 Mag 4 against its 19.5 km unless stated. The robust kernel is not sufficient to explain the failure: removing it entirely gives 33.0 km, and widening the Huber constant from 1.345 to 10 and 100 gives 18.0 and 33.0 km, while Mag 5 of the same line returns 11.7 m with the kernel and 11.7 m without it. Map resolution is not sufficient: re-exporting the anomaly grid at 34 m spacing instead of 113 m gives 13.7 km. The lag is not sufficient: at $\sigma_\beta = 1$, lags of 300, 450, 600 and 900 s give 19.5, 14.8, 14.7 and 14.5 km. Retained linearization is not sufficient: relinearizing every variable at every update gives 9.8 km. Measurement thinning is not the cause: attaching all ten raw samples to their nearest kept state gives 22.9 km. Nor is the measurement noise, which is the alternative worth taking most seriously because the compensated residual on Mag 4 is several times the assumed $\sqrt{R} = 12$ nT: no single value of $\sqrt{R}$ repairs it across lines. Setting it to 150 nT recovers 1007.06 to 64.5 m and leaves 1003.02 at 31 km; setting it to 65 nT recovers 1007.02 to 41.8 m and leaves 1007.06 at 7.3 km. Across the ten runs the result ranges from 42 m to 46 km with no setting bounded everywhere, where raising σ_β is bounded on every line at once. The 10^{-20} conditioning floor on the process noise is likewise not doing the work: raising it to 10^{-16} gives 49 km and lowering it to 10^{-24} gives 12.9 km. Each of these moves

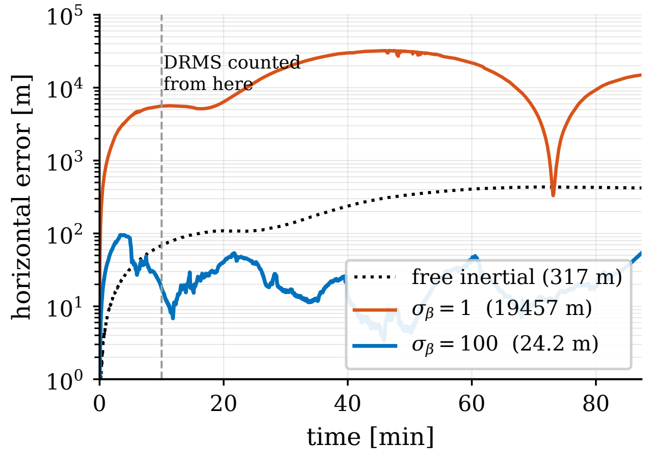

Fig. 16. Per-epoch smoother on line 1007.06 Mag 4 at a 300 s lag. At the nominal compensation prior the estimate leaves the map in the first minutes and never returns; with the compensation prior raised to $\sigma_\beta = 100$, and nothing else changed, it stays an order of magnitude below free-inertial drift for the whole line.

the number and none recovers the solution, which is what separates a contributing factor from the one setting that does.

Two measurements bound how far the modeling error reaches. The same graph for line 1007.06 Mag 4, solved as one full-line batch from the same zero start and the same nominal prior, reaches 18.6 m: over a whole line the compensation has time to move and the map has gradient enough to hold position, so the cost function itself is sound and the sensitivity is a property of short spans. The sliding window of Section B, run in this same code base, sits between the two. At the nominal prior it diverges on 1007.06 Mag 4 (5430 m) but not on 1003.02 Mag 4 (36.6 m), and seeding its first window with a ridge fit of the compensation brings 1007.06 to 30.8 m, which is the cross-implementation comparison of Section V-F. The sensitivity is therefore not peculiar to per-epoch marginalization; it grows as the span each solve sees gets shorter.

A caveat on the divergent runs themselves. The exported anomaly grid covers the trajectory with a 2.2 km margin, and the interpolator clamps outside it, so once an estimate leaves the grid the map gradient vanishes and no information can pull it back. The kilometre-scale numbers in the first column of Table V are therefore a floor on how bad the divergence is, not a measurement of where an unclamped estimator would have gone; what they establish is that the estimate left the map, not how far.

I. The same prior, seen in a single window

The explanation can be tested one window at a time, which also answers how much flight the joint problem needs before it is determined. Table VII reports the solution reached when the same graph is solved as a

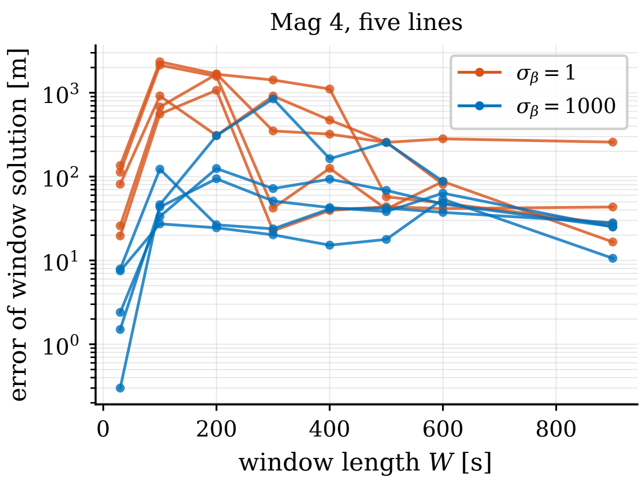

Fig. 17. Position error of the batch solution over the first W seconds of each Mag 4 line, from a zero compensation. At the nominal prior the solutions climb to kilometres over the middle window lengths; at the widened prior the excursion is absent on every line but the short calibration line 1006.08.

batch over only its first W seconds, from a zero compensation, at both priors, with no commitment and no marginalization anywhere. At $\sigma_\beta = 1$, the Mag 4 solutions pass through a region, covering at least $100\,\mathrm{s}$ to $200\,\mathrm{s}$ on all five lines and extending to $400\,\mathrm{s}$ on 1007.06, in which they sit hundreds of metres to kilometres from truth, while Mag 5 stays within tens of metres except on the short line 1006.08. At $\sigma_\beta = 1000$, that region is gone: every Mag 4 entry outside 1006.08 falls between 2 and $125\,\mathrm{m}$, and line 1007.02 Mag 4, which at the tight prior never left the region within the sweep, is at or below $94\,\mathrm{m}$ at every window length. The single-window and the full-line experiments therefore agree, and they turn on the same parameter (Fig. 17).

This also fixes what the failure needs. It is not early commitment: these window solves commit nothing and still land kilometres out. It is a short data span together with a prior that is not scaled to the regressor. Marginalization only decides whether the estimator can recover afterwards, and in the fixed-lag case it cannot.

Line 1006.08 is the exception at both priors. It stays at hundreds of metres across its middle window lengths and improves only slowly, which is consistent with the low map-information floor of Remark 2 rather than with the compensation: where $\mathbf{G} \approx \mathbf{0}$, no prior on $\boldsymbol{\beta}$ can help.

One negative result belongs here because we specifically tested for it. We also repeated the sweep from a compensation seeded with a ridge fit of $(z_t - h(\mathbf{p}_t^{\mathrm{ins}}))$ over the same W seconds, to test whether the far-from-truth solutions are a matter of which minimum the solver selects. Across the 78 combinations of line, magnetometer, and W, the two starts reach objectives that differ at all in only 7; of those, the seeded start is the cheaper and the more accurate one in 5, and in 2 the cheaper solution is the less accurate one. The window solve is largely indifferent to where it starts. We therefore do not attribute the cold-start failure to minimum selection, and we do not claim that the accurate solution can be identified online by comparing objective values.

J. Real-time cost

With the compensation prior raised to $\sigma_\beta = 100$, the per-epoch smoother runs the full breadth at a $300\,\mathrm{s}$ lag. Graph states are placed at $1\,\mathrm{Hz}$ rather than at the $10\,\mathrm{Hz}$ sample rate, with the transition compounded and the process noise accumulated exactly over the ten intervening samples, so the $300\,\mathrm{s}$ lag is a 300-state window updated once a second. Across the ten cases, the update takes $6.0\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state, mean $7.8\,\mathrm{ms}$, so an $87\,\mathrm{min}$ line finishes in $42\,\mathrm{s}$ and the margin against the $1\,\mathrm{Hz}$ state cadence is two orders of magnitude. These are mean times on one core of a laptop-class x86 processor under WSL2, with the measurement factor evaluated through a Python callback, so they are an upper bound on what native factors would cost; the corresponding costs in the relinearization and dense-measurement runs are $75.5\,\mathrm{ms}$ per state with every variable relinearized at every update and $28.7\,\mathrm{ms}$ to $35.1\,\mathrm{ms}$ with all ten raw measurements attached. Placing states at the full $10\,\mathrm{Hz}$ instead makes the same lag a 3000-state window, which is why a $300\,\mathrm{s}$ lag is reported here at $1\,\mathrm{Hz}$ states and not at the sample rate.

The decimation costs the nine measurements dropped between kept states, and on these lines it does not cost accuracy. The aircraft covers $68\,\mathrm{m}$ per kept state at the survey ground speed. Attaching all ten raw samples to their nearest kept state instead of one returns $26.7\,\mathrm{m}$ against $23.9\,\mathrm{m}$ on line 1007.06 Mag 4 and $26.2\,\mathrm{m}$ against $19.2\,\mathrm{m}$ on 1003.02 Mag 4, both pairs run with the ridge-seeded cold start rather than the widened prior because that is the pair we have, for 3.5 to 3.8 times the computation. Two things are folded into that comparison and we can separate neither with the runs we have: the dense variant attaches all ten factors to one state, so it also assumes a single error state over the $1\,\mathrm{s}$ span, and the measurements themselves are unlikely to be independent at $10\,\mathrm{Hz}$ given that the compensated residual on these lines is several times $\sqrt{R}$. We report the comparison as a reason not to treat the decimation as a pure loss, not as evidence about the noise model, which would need an innovation whiteness test we have not run on these lines.

These same properties make the estimator compatible with online, fixed-lag operation rather than zero-latency filtering. Its committed output lags real time by the overlap L_o, here $90\,\mathrm{s}$, a latency that suits the map-aiding role, in which the INS supplies the high-rate real-time solution and the smoother corrects it periodically at the commit cadence ($L_w - L_o$, here $3.5\,\mathrm{min}$), and that the designer trades against accuracy through L_o. The factor interface is modular, so compensation and map factors can be added or removed without touching the solver, and one implementation then covers both the calibrated case with

TABLE V

Per-epoch incremental smoother, $300\,\mathrm{s}$ lag, $1\,\mathrm{Hz}$ states, at the nominal compensation prior and at the recommended one. Smoothed DRMS [m] after a $600\,\mathrm{s}$ warm-up, causal value in parentheses; DRMS is evaluated at the $1\,\mathrm{Hz}$ state grid, and INS is the free-inertial drift computed on the same grid by the same code. "div." $> 1\,\mathrm{km}$ and, given the grid clamping discussed in the text, is a lower bound. Line 1006.08 is the $14\,\mathrm{min}$ calibration line set aside in Section V; on its Mag 5 the smoothed estimate is worse than the causal one at both priors, which we read with Remark 2 as the map-information floor rather than as a smoothing gain.

Line	Mag	INS	$\sigma_\beta = 1$	$\sigma_\beta = 100$
1003.02	4	123.8	div.	**19.6** (58.2)
1003.02	5	123.8	12.6 (24.4)	12.0 (21.9)
1003.08	4	271.9	div.	**25.0** (48.9)
1003.08	5	271.9	10.9 (22.4)	10.5 (19.2)
1006.08	4	197.7	div.	**99.1** (134.8)
1006.08	5	197.7	83.9 (52.4)	101.0 (80.1)
1007.02	4	120.7	60.0 (327.8)	**28.4** (74.3)
1007.02	5	120.7	14.0 (40.3)	15.8 (33.1)
1007.06	4	317.5	div.	**24.2** (42.7)
1007.06	5	317.5	11.7 (19.8)	11.5 (16.0)

TABLE VI

Prior sweep on the Mag 4 lines, zero prior mean, no other change: smoothed DRMS [m] after a $600\,\mathrm{s}$ warm-up. The first four columns scale the single shared coefficient standard deviation σ_β; "per col." instead gives every regressor column a prior of $100\,\mathrm{nT}$ of measurement contribution; "σ_S" leaves the compensation prior at nominal and widens the FOGM disturbance prior and walk a hundredfold instead.

	σ_β				per col.	σ_S
Line	1	10	100	1000	100 nT	$\times 100$
1003.02	26810	32.7	19.6	19.3	**19.0**	33.2
1003.08	11262	25.7	25.0	24.5	**25.1**	29.8
1006.08	6574	117.7	99.1	85.4	**98.9**	1144
1007.02	60.0	30.3	28.4	30.3	**30.5**	56.7
1007.06	19457	24.1	24.2	23.9	**23.7**	27.3

TABLE VII

Solution reached by Levenberg–Marquardt from a zero compensation on a batch over only the first W seconds of each line: horizontal DRMS [m] over those W seconds, at the nominal and at a uniformly widened compensation prior. These are the solutions the solver reaches, not certified global optima. The free-inertial drift over the same spans runs from $0.4\,\mathrm{m}$ to $5.0\,\mathrm{m}$ at $W = 30\,\mathrm{s}$ to $56.5\,\mathrm{m}$ to $100.7\,\mathrm{m}$ at $W = 900\,\mathrm{s}$, so an entry below about $100\,\mathrm{m}$ is at or under the drift it has to beat.

			W [s]							
Line	Mag	σ_β	30	100	200	300	400	500	600	900
1003.02	4	1	113	2137	1569	42	126	41	87	17
		1000	8	27	24	20	15	18	54	11
1003.02	5	1	11	42	13	10	9	24	8	11
		1000	12	55	11	12	11	14	16	11
1003.08	4	1	20	558	1069	23	39	44	42	43
		1000	8	123	27	24	41	42	37	28
1003.08	5	1	26	70	14	12	24	20	24	22
		1000	3	18	11	3	12	15	11	10
1006.08	4	1	81	917	310	915	472	257	81	–
		1000	<1	46	308	847	164	255	88	–
1006.08	5	1	48	136	341	70	115	50	36	–
		1000	6	67	113	35	81	26	31	–
1007.02	4	1	136	2344	1676	350	321	256	282	258
		1000	2	43	94	51	43	38	63	25
1007.02	5	1	13	15	32	34	44	34	60	34
		1000	3	18	22	23	41	37	66	35
1007.06	4	1	26	676	1662	1420	1109	57	48	27
		1000	2	33	125	72	93	68	47	25
1007.06	5	1	23	78	90	20	17	19	19	12
		1000	1	8	26	11	13	12	10	7

frozen coefficients and the cold-start case with adaptive coefficients. Integrating it with an operational system amounts to feeding the INS mechanization, fluxgate, and scalar magnetometer streams into the graph and reading back the corrected position at the commit rate. The main tuning burden is the two window parameters and the compensation process noise $\mathbf{Q}^\beta$, all of which have clear physical meaning.

VI. Conclusion

This paper formulated airborne magnetic-anomaly navigation as factor-graph MAP estimation and used the formulation to solve the cold-start compensation problem jointly with navigation. The Tolles–Lawson coefficients enter the graph as time-varying variables, so the compensation adapts along the flight and late-arriving calibration information corrects early navigation states, the mechanism a one-pass causal filter lacks; and a linear condition states when position and compensation can be separated at all. That single MAP problem is realized across a lag spectrum, from a full-line batch that fixes the offline accuracy bound, through a fixed-lag window, to a per-epoch incremental smoother that runs a $300\,\mathrm{s}$ lag in real time at $6.0\,\mathrm{ms}$ to $9.0\,\mathrm{ms}$ per state. On real SGL 2020 data the estimator stays bounded from a cold start on all eight full-length cabin-magnetometer cases, where the plain online-TL EKF diverges and a marginalized particle filter is erratic, and it is more accurate on every such case than the tuned online EKF+TL+NN baseline, with no neural network and no calibration flight. A Monte-Carlo simulation shows its reported covariance is consistent and conservative where the toolbox particle filter's is over-confident; the recorded platform currents confirm the absorption mechanism, explaining a third of the raw cabin interference but almost none of the post-fit residual; and an independent GTSAM reimplementation that shares no code reproduces the bounded cold-start result, separating the estimator from its implementation.

Several limitations bound the study. Each real line is a single physical realization, so its DRMS is a point estimate, and the cross-implementation check is likewise a single run that establishes implementation independence rather than a statistical claim; distributional statements come from the Monte-Carlo simulation only. The baseline set, though it now spans a weak causal filter, the strongest reported causal method, and a particle filter, still lacks a grid or point-mass MMSE map-matcher [21], [8]. The compensation basis is attitude-only: physically structured sensor-error states (heading harmonics, dead zones) and direct regression of current telemetry are natural graph extensions, but our exploratory tests show that applying either blindly, without documented sensor hardware and orientation, or with naïvely scaled telemetry, can help or hurt by tens of percent depending on the line, so neither is claimed here. The warm-start prior used above is drawn from the same line rather than transferred from a separate calibration flight, so a true cross-line warm start remains untested, and the observability gate the separability condition suggests is analyzed but not yet folded into the online estimator. The per-epoch smoother reaches a $300\,\mathrm{s}$ lag in real time only by placing states at $1\,\mathrm{Hz}$; at the full $10\,\mathrm{Hz}$ state cadence, that lag is a window ten times as large and remains offline in this implementation, a limit we expect native factors to lift. The compensation prior the cold start needs is a measured setting, not a derived one: a shared $\sigma_\beta = 1$ is far too tight, anything from 10 to 1000 works on these installations, and a dimensionally consistent per-column prior works equally well, but we give no rule for choosing the scale from installation data, and no setting we found both fixes the cold start and holds the coefficients at the magnitude a direct fit of the same data needs. A related limit bounds the comparison itself. The window and filter results of Section V were produced at the nominal prior. The window smoother is bounded there on every counted case, so its numbers are not themselves an artifact of the prior, though the independent GTSAM window diverges on 1007.06 Mag 4 at that same prior (Section V-F), so the boundedness is not robust across implementations. We have not re-run either the window or the causal EKF baselines at a corrected prior. Part of the EKF cold-start divergence on the cabin magnetometers may therefore be the same modeling effect rather than a property of causal filtering, and the margin the window shows over those baselines is an upper bound on the margin that survives a corrected prior. And none of the SGL 2020 lines is a long, straight-and-level operational transit: the attitude content that stresses compensation also excites the TL regressor and aids its observability, so performance under the weak excitation of a pure cruise is not established here. Short lines remain hard for every method, an information limit rather than an estimator one.

These limitations set the agenda for further work: extending the evaluation to the full SGL line set and a grid or point-mass baseline; lag scheduling that runs a long lag through the early, weakly observable minutes and contracts once the compensation has converged, moved online by native factors; transferring a compensation prior across lines for a true warm start; folding the separability condition into an online observability gate; making the sensor-error and current-telemetry extensions robust enough to claim, both staying within the attitude-derived, position-independent basis the condition requires; and testing a low-excitation operational cruise. None of these qualifications affects the central and reproducible finding: carrying the compensation as time-varying variables of a windowed factor graph, solved anywhere along the lag spectrum, is sufficient to survive the cold start and to match or beat an NN-augmented causal filter on this benchmark, with every estimated quantity a physically interpretable state.

Appendix A
Pinson Error-State Dynamics

The navigation-error block of $\mathbf{\Phi}_t$ in (2) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors $(\delta\mathbf{p}, \delta\mathbf{v}, \boldsymbol{\psi})$ and sensor biases $(\mathbf{b}_a, \mathbf{b}_g)$, the continuous error dynamics are

$$\begin{aligned} \delta\dot{\mathbf{p}} &= \mathbf{F}_{pp}\,\delta\mathbf{p} + \delta\mathbf{v}, \\ \delta\dot{\mathbf{v}} &= \mathbf{F}_{vp}\,\delta\mathbf{p} + \mathbf{F}_{vv}\,\delta\mathbf{v} - [\mathbf{f}^n\times]\,\boldsymbol{\psi} + \mathbf{C}_b^n\mathbf{b}_a, \\ \dot{\boldsymbol{\psi}} &= \mathbf{F}_{\psi p}\,\delta\mathbf{p} + \mathbf{F}_{\psi v}\,\delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n\times]\,\boldsymbol{\psi} - \mathbf{C}_b^n\mathbf{b}_g, \end{aligned} \tag{13}$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, $\boldsymbol{\omega}_{in}^n$ the inertial rate, $[\cdot\times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ. The discrete $\mathbf{\Phi}_t = \exp(\mathbf{F}_t\,\Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [44]. Only $\mathbf{g}_t^\top$, the position row of the measurement Jacobian (10), couples this block to the map.

Appendix B
Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of (6) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [\,u\ v\ w\,]^\top$ (with $u^2+v^2+w^2=1$) and the scalar field magnitude B [11], [12]. The permanent moment contributes the three cosines (u, v, w); the induced magnetization, symmetric in the field, contributes the six products $B\,(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B\,(u\dot{u}, u\dot{v}, \ldots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top\boldsymbol{\beta}_t$ is linear in the coefficients $\boldsymbol{\beta}_t$, the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes $\boldsymbol{\beta}$ is unavailable, which is precisely why we carry it as a graph variable.

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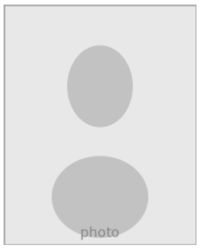
Jayden Dongwoo Lee received the M.S. degree in aerospace engineering from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2021. He is currently pursuing the Ph.D. degree in aerospace engineering at KAIST. His research interests include unmanned aircraft systems (UAS), guidance and navigation, adaptive and data-driven control, and estimation.



Bohyun Chang is with the Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea. His research interests include guidance, navigation, and control of aerospace vehicles. (Detailed biography to be provided by the author.)

Hyochoong Bang received his Ph.D. degree in aerospace engineering from Texas A&M University in 1992. Since 2001, he has been with the Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, as a professor. His research interests include spacecraft guidance, attitude dynamics and control, unmanned aerial vehicle flight dynamics and control, and flight control system design.

Appendix C
Proof of Proposition 1

Two joint states $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ and $(\delta\mathbf{p}', \delta\boldsymbol{\beta}')$ produce identical noise-free predictions iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \boldsymbol{\Psi}(\delta\boldsymbol{\beta}' - \delta\boldsymbol{\beta})$, i.e. iff $[\,\mathbf{G} \;\; \boldsymbol{\Psi}\,][(\delta\mathbf{p} - \delta\mathbf{p}')^\top \; (\delta\boldsymbol{\beta} - \delta\boldsymbol{\beta}')^\top]^\top = \mathbf{0}$. The unobservable subspace is therefore exactly $\ker[\,\mathbf{G} \;\; \boldsymbol{\Psi}\,]$, and the joint state is observable iff $[\,\mathbf{G} \;\; \boldsymbol{\Psi}\,]$ has full column rank; the first statement of the proposition follows by taking $\delta\boldsymbol{\beta}'$ free and $\delta\mathbf{p}' = \mathbf{0}$.

For the three-condition equivalence: if (i) fails, take $\delta\mathbf{p} \in \ker\mathbf{G} \setminus \{\mathbf{0}\}$, giving the kernel element $(\delta\mathbf{p}, \mathbf{0})$; if (ii) fails, take $\delta\boldsymbol{\beta} \in \ker\boldsymbol{\Psi} \setminus \{\mathbf{0}\}$, giving $(\mathbf{0}, \delta\boldsymbol{\beta})$; if (iii) fails, take $\mathbf{G}\,\delta\mathbf{p} = \boldsymbol{\Psi}\,\delta\boldsymbol{\beta} \neq \mathbf{0}$, giving $(\delta\mathbf{p}, -\delta\boldsymbol{\beta})$. Conversely, let $(\delta\mathbf{p}, \delta\boldsymbol{\beta}) \neq (\mathbf{0}, \mathbf{0})$ lie in the kernel, so $\mathbf{G}\,\delta\mathbf{p} = -\boldsymbol{\Psi}\,\delta\boldsymbol{\beta}$. If $\mathbf{G}\,\delta\mathbf{p} = \mathbf{0}$ then either $\delta\mathbf{p} \neq \mathbf{0}$ violates (i) or $\delta\boldsymbol{\beta} \neq \mathbf{0}$ violates (ii); if $\mathbf{G}\,\delta\mathbf{p} \neq \mathbf{0}$ it is a nonzero shared direction, violating (iii). ■

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